

ACP 245 – Application Communication Protocol

**Prepared by
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DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Revision History

Revision	Date	Author	Description
1.2	17/02/2009	DENATRAN	Item 2.1 Rules is added.
1.2	17/02/2009	DENATRAN	A note is added under the table 1 of Item 2.5
1.2	17/02/2009	DENATRAN	Value 32 is added in Item 3.1.2
1.2	17/02/2009	DENATRAN	Value 135 is added in Item 3.1.3
1.2	17/02/2009	DENATRAN	Auth. Key field is included in the structure of Item 3.4
1.2	17/02/2009	DENATRAN	Table from Item 3.4.1 is updated
1.2	17/02/2009	DENATRAN	Table from Item 3.5.1 is updated
1.2	17/02/2009	DENATRAN	A comment for value = 0 is added in item 3.8.5
1.2	17/02/2009	DENATRAN	Table from Item 3.8.10 is updated
1.2	17/02/2009	DENATRAN	A comment is added under structure of Item 3.9 and third flag in Item 3.9.1 is modified.
1.2	17/02/2009	DENATRAN	A comment is added in item 4.1.2 Private Flag
1.2	17/02/2009	DENATRAN	Message Type table is added in Item 5.1
1.2	17/02/2009	DENATRAN	Grace Time is added to structure of Item 5.2 and Item 5.2.1.6
1.2	17/02/2009	DENATRAN	Comments are added to Items 5.2.1.4 and 5.2.1.5
1.2	17/02/2009	DENATRAN	Message Type table is added in Item 6.1
1.2	17/02/2009	DENATRAN	Text is updated in Item 7.
1.2	17/02/2009	DENATRAN	Message Type table is added in Item 7.1
1.2	17/02/2009	DENATRAN	The structure of Item 7.3 is updated according ACP v. 3.1.0.2
1.2	17/02/2009	DENATRAN	Message Type table is added in Item 8.1
1.2	17/02/2009	DENATRAN	Text is updated in Bit 1 of Item 8.4.1.3.1
1.2	17/02/2009	DENATRAN	Message Type table is added in Item 9.1
1.2	17/02/2009	DENATRAN	Mandatory parameters are defined in Appendix I.
1.2	19/02/2009	DENATRAN	Removed "Raw Data" table from Item 3.7.2.
1.2	19/02/2009	DENATRAN	Keep Alive Timer in Item 12.1.5 is changed to 180 seconds.
1.2	19/02/2009	DENATRAN	The "Type" of APN, Login and Password is merged to max. 90 Bytes in Item 12.1.6
1.2	19/02/2009	DENATRAN	Example 1, 2 and 3 in Appendix II are revised.
1.2	06/03/2009	DENATRAN	Added new Data Type to TCU Service Activation/ Deactivation
1.2	06/03/2009	DENATRAN	Changed all the Auth key sizes from 8 bytes to undefined
1.2	06/03/2009	DENATRAN	Item 3.8 – Inserted the location element structure according to ACP 3.1.0.2
1.2	06/03/2009	DENATRAN	Item 2.1 – Removed “optional” word
1.2	06/03/2009	DENATRAN	Item 3.4 – “Simcard ID” and “Auth key” order changed
1.2	06/03/2009	DENATRAN	Item 3.6.1 - Removed “vehicle blocking” and vehicle unblocking”. Added “Anti theft blocking (ACP245)”
1.2	06/03/2009	DENATRAN	Item 5.2.1.3.1 – Added target Application ID as 0 to mandatory configuration parameters
1.2	06/03/2009	DENATRAN	Item 9.4 – Added vehicle descriptor element to the message structure
1.2	06/03/2009	DENATRAN	Item 12.1.1 – Changed default values to 300
1.2	06/03/2009	DENATRAN	Item 12.1.5 – Changed type index 0x0051 type to unsigned int/2 bytes. Added definition for value 0

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1.2	06/03/2009	DENATRAN	Item 12.1.6 – fixed data type indexes from 0x0061 to 0x0063 type to “Max 91 bytes (sum)”
1.2	06/03/2009	DENATRAN	Changed server ports types to 2 bytes
1.2	06/03/2009	DENATRAN	Item 13.2.1, 13.2.2, 13.3.1 - Fixed example
1.2	23/03/2009	DENATRAN	Values from 33 to 61 are added in Item 3.1.2
1.2	09/04/2009	DENATRAN	Values from 61 to 63 are added in Item 3.1.2
1.2	09/04/2009	DENATRAN	Added remarks in Items 4.1.7, 5.1, 6.1, 7.1, 8.1, 8.4, 9.3, 9.6
1.2	09/04/2009	DENATRAN	Fixed Figure 2 in Item 6
1.2	09/04/2009	DENATRAN	Item 12.1.1 – Default unit from Data type 0x0012 changed to “Minutes”
1.2	09/04/2009	DENATRAN	Fixed message structure in item 6.3 adding “Reserved” element
1.2	09/04/2009	DENATRAN	Figure 4.1 is added in item 8
1.2.1	17/04/2009	DENATRAN	Item 3.6.1 – Added remark regarding Entity ID = 17.
1.2.1	17/04/2009	DENATRAN	Items 3.8.8 and 3.8.9 – Added remark about 2's complement.
1.2.1	17/04/2009	DENATRAN	Item 4 – Added “optional” in octet 4
1.2.1	17/04/2009	DENATRAN	Item 4.1.6 – Added remark.
1.2.1	17/04/2009	DENATRAN	Items 5.2, 6.2, 6.5, 7.2 – “(bit 3=1)” is added in Message Control Flag.
1.2.1	17/04/2009	DENATRAN	Items 5.2.1.1, 6.2.1.1, 6.5.1.1 and 7.2.1.1 – Added remark regarding Message Control Flag.
1.2.1	17/04/2009	DENATRAN	Item 5.2.1.3.1 – Fixed value for Target Application ID in messages Appl= 2 Message type=8.
1.2.1	17/04/2009	DENATRAN	Item 6.2.1.3 – Added remark regarding Target Application ID = 0
1.2.1	17/04/2009	DENATRAN	Item 8.2, Message type =1 is removed.
1.2.1	17/04/2009	DENATRAN	Item 8.4.1.3 – Defined default values.
1.2.1	17/04/2009	DENATRAN	Item 9 – Figure 5 is updated.
1.2.1	17/04/2009	DENATRAN	Item 9.2 – Added remark.
1.2.1	17/04/2009	DENATRAN	Item 9.4 is changed. “Vehicle Position Message” is added.
1.2.1	17/04/2009	DENATRAN	Item 9.6 – Added remark.
1.2.1	17/04/2009	DENATRAN	Item 12.1.1 – Fixed default value for “Tracking Timer” parameter.
1.2.1	17/04/2009	DENATRAN	Item 12.1.10 – Added custom parameters
1.2.1	17/04/2009	DENATRAN	Item 13.3.1 – Added position coordinates in degrees, minutes, seconds
1.2.1	17/04/2009	DENATRAN	Item 13.4 – Added example to convert position coordinates
1.2.1	22/04/2009	DENATRAN	Item 6.2 – Fixed text “Message Fields” element in structure.
1.2.1	22/04/2009	DENATRAN	Item 13.2.2 – Fixed order of elements in message structure.
1.2.1	23/04/2009	DENATRAN	Item 13.2.1 – Fixed value in message string.
1.2.1	30/04/2009	DENATRAN	Item 14 – Added Appendix III
1.2.1	30/04/2009	DENATRAN	Item 3.1.3 – Added new TCU manufacturer
1.2.1	06/05/2009	DENATRAN	Added the mapping of bits – item 4.1.7
1.2.1	06/05/2009	DENATRAN	Added the mapping of bits – item 5.2.1.3.3 e item 5.2.1.3.4
1.2.1	06/05/2009	DENATRAN	Added the mapping of bits – item 8.4.1.3.1 e item 8.4.1.3.3
1.2.2	05/08/2009	DENATRAN	Added data type 0x0047 on item 12.1.4
1.2.2	05/08/2009	DENATRAN	Updated items 14.6, 14.8 on Appendix III
1.2.2	05/08/2009	DENATRAN	Added remark on item 6.5.1.4.5

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1.2.2	05/08/2009	DENATRAN	Added remark in item 12.1.1 (data type 0x0011)
1.2.2	05/08/2009	DENATRAN	Added remark in item 3.8
1.2.2	05/08/2009	DENATRAN	Updated title in item 14.11.2
1.2.2	07/08/2009	DENATRAN	Added item 14.9, 14.10, 14.11 on Appendix III
1.2.2	07/08/2009	DENATRAN	Item 13 – Appendix II was revised.
1.2.2	07/08/2009	DENATRAN	Added remark in item 2.1.
1.2.2	07/08/2009	DENATRAN	Added remark in items 3.7.1, 3.9.1, 3.9.3.
1.2.2	07/08/2009	DENATRAN	Value 99 and 127 are added in Item 3.1.2
1.2.2	18/08/2009	DENATRAN	Value 65 is added in Item 3.1.2
1.2.2	27/08/2009	DENATRAN	Added new table and Description in item 4.1.6 to represent the Versions of ACP245
1.2.2	27/08/2009	DENATRAN	Added Item 4.1.11, new extended Version Number of ACP 245
1.2.2	27/08/2009	DENATRAN	Added example of Extended Version Number of ACP245 in Item 13.7.1.6
1.2.2	18/08/2009	DENATRAN	Value 66 is added in Item 3.1.2

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New Features

Feature	Date	Author	Description
1.2	17/02/2009	DENATRAN	Entity Id = 17 is added in the table of Item 3.6.1
1.2	17/02/2009	DENATRAN	Added HDOP value in Item 3.8.9 to represent position uncertainty estimate.
1.2	17/02/2009	DENATRAN	It is added the possibility of sending status of breakdown source using breakdown data in Item 3.9.2 (Bit 7) & Item 3.9.3
1.2	17/02/2009	DENATRAN	TCU data element modified to allow multiple configuration parameters
1.2	17/02/2009	DENATRAN	Added TCU data error element in item 3.12 to allow reply multiple configuration parameters
1.2	17/02/2009	DENATRAN	Added new message types ID=8 e 9 to allow multiple configurations and ID=10 for SMS message in item 6.1
1.2	17/02/2009	DENATRAN	Added new configuration reply message #2 ACP 245 ID=9 in Item 6.4
1.2	17/02/2009	DENATRAN	Added new message types ID=4 e 5 to allow keep alive messages in items 9.4 and 9.5
1.2	19/02/2009	DENATRAN	Added Auth.Key field in Item 3.4 Vehicle Descriptor Element
1.2	19/02/2009	DENATRAN	Added new error element 27, 28 and 29 in Item 3.5.1
1.2	19/02/2009	DENATRAN	Section 10 is added.

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ACRONYMS

The following acronyms apply to this document.

Acronym	Definition
ACP	Application Communications Protocol
DENATRAN	National Traffic Department
FOTA	Firmware Over The Air
GMT	Greenwich Mean Time
GPRS	General Packet Radio Service
GPS	Global Positioning System
GSM	Global System for Mobile Communication
IE	Information Element
IMEI	International Mobile Equipment Identity
LSB	Least Significant Bit
MSB	Most Significant Bit
SIM CARD	Subscriber Identity Module
SO	Service Operator
TCP	Transmission Control Protocol
TCU	Telematics Control Unit
UDP	User Datagram Protocol
UTC	Coordinate Universal Time

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1 INTRODUCTION

This document was based on Application Communication Protocol (ACP), developed by Motorola Inc. (version 3.1.0.2 / November, 2000 and made public in the year of 2002) and describes the basic ACP services to use in tracking tracing systems, in agreement with DENATRAN regulation. It suggests the flow messages on each ACP service as well correspondent composition and minimal message structure elements. Tracking device and Service Operator are responsible for exchange messages and keeping available services.

1.1 Telematic Applications

The Basic telematics services IDs. Table 1 presents the basic, but not limited, telematic applications in accordance with DENATRAN regulation. They have the purpose to group different messages, originating the idea of services and context.

Application ID	Definition	Brief description
1	Provisioning	Basic messages whereby the TCU establishes and manages configurations for applications required for activation.
2	Configuration	Application configurations that are not handled by the generic provisioning service.
6	Remote Vehicle Function	This application allows control functions to occur within the vehicle. Examples are remote door unlock.
10	Vehicle Tracking	This application provides the ability to track a vehicle, usually due to theft.
11	Alarm	This application provides the ability to recognize unauthorized starting or movement of a vehicle.

Table 1. Basic telematics services IDs.

This document describes in section 2 brief rules about the data used in each information element into the messages. It follows with a description of elements and data structures that will composite an ACP message.

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2 Data Description

This section describes the data format used in ACP Protocol for message exchange between TCU and SO. The information bellow was extracted from the ACP Specification [1].

2.1 Rules

This section describes the message formats used in the exchange of data between a TCU and the SO. Message formats consist of information elements as needed to create a meaningful message. This standard defines the minimal set of messages necessary to support the telematics functions. Undefined and non-supported messages and information elements can be skipped. The intent is to allow for the addition of messages information elements without affecting existing applications of current customers. This rule is intended to promote robustness and is not intended to encourage sending useless information between TCU and SO.

The basic unit of data is an information element. Information elements can be combined to form message types, the unit used for the transfer of data by the transport level. A number of message types are defined in this standard along with their use.

The information element is position dependent in the message structure. Elements that do not contain data must still be included. Elements may not be included in one of three ways:

1. The length field and More Flag of the element is zero,
2. A control flag is used to indicate the presence or absence of an element,
3. The length field of the parent element does not include the element.

The Header Element is not the parent element of the whole message.

2.2 Bit Order

Bits in an octet are numbered from 0 to 7 from most significant bit (MSB) to least significant bit (LSB). Multi-octet fields have the most significant octet in the in the first octet of the field.

2.3 More Flag

This value identifies if the last information in the octet was not finished and will be continue in the next octet. The value 0 means no more information, while the value 1 means more data.

Octet / Bit	0	1	2	3	4	5	6	7
1	IE Identifier		More Flag=1	Length ₀ = 2				
2	More Flag=0	Length ₁ = 4						

Table 2. More flag sample

In this example, presented by table 2, the field Length in the first octet continues in the second octet because More Flag is set with the value 1. It means Length is composed by Length₀ (MSB) and Length₁ (LSB).

2.4 Additional Flag

This concept is the same of More Flag. The additional flag (Addl Flag) is used to indicate when an element of like kind is repeated in the following octet. An example would be status flags pertaining to GPS status in Field₀ would have additional status flags pertaining to GPS in Field₂ and Field₄.as showed in table 3.

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Octet / Bit	0	1	2	3	4	5	6	7
1	Addl Flag=1	Field ₀			Field ₁			
2	Addl Flag=1	Field ₂			Field ₃			
3	Addl Flag=0	Field ₄			Field ₅			

Table 3. More Flag sample

2.5 Element Definition

The information element identifies a block of data. The initial octet has three fields, respectively: Identification Element (IE) – 2 bit, More Flag – 1 bit and Length (data size of block). The more flag is related with the Length field. If the length field value exceeds 32 octets of data (0..31 - 5 bits) then more flag must be set 1 and another Length field follows. The final value is composed by Length₀ (5 bits) and Length₁ (7 bits) – 12 bits; as in the example bellow.

Octet / Bit	0	1	2	3	4	5	6	7
1	IE Identifier = a		More Flag = 1	Length ₀ = x				
2	More Flag=0	Length ₁ = y						

The valid values for IE are:

Value	Definition	
0	Transparent data (binary)	Number of 8 bit octets
1	Text (ISO 8859-1) / 8bits per character	Number of 8 bit octets
2	Packed decimal	Number of 8 bit octets
3	Extended	Extended definition

Table 1 - Default IE (Identification Element) values.

Example: a = 0, x = 2 and y = 4 - **Binary** data with **Length** = 000010 (Length₀) 0000100 (Length₁) = 0000100000100₂ = 260.

The ACP 245 must be fully implemented by the SO's. The TCU's will use the option IE values that are for alphanumeric data = 1 (Text (ISO 8859-1) / 8bits per character) and for numeric field = 2 (Packed decimal/BCD)

If IE value is *Extended*, the initial structure of the element will be changed. The first bit of Length₀ field starts at the next information octet.

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Octet \ bit	0	1	2	3	4	5	6	7
1	IE = 3 (Extended string definition)		More Flag = 1	Extended IE identifier (0..31) = a				
2	More Flag = 0	Length ₀ (number of octets that follow) (0..127) = x						
3...n-1	Data = b							

The valid values for Extended IE are:

Value (Decimal)	Definition
0	Transparent data (binary)
1	Text (ISO 8859-1) / 8bits per character
2	Packed decimal
3	Reserved
4	UNICODE
5	UTF-8
6	Shift JIS
7..30	Reserved
31 *	Extended Private Identifier

Table 2 - Extended IE (Identification Element) values.

* The value of 31 indicates that the element is not defined in this document and is considered proprietary or private.

Example: a = 1, x = 20, b = “Test of ACP Protocol” - Text data with Length = 00010100₂ = 20.

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3 Message Elements

All the messages are composed by elements -information units. Find bellow examples of common elements.

3.1 Version Element

Define information about manufacturers of the vehicle and TCU.

Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Version Element	IE Identifier = 0		More Flag	Length				
	Car Manufacturer ID							
	TCU Manufacturer							
	Major hardware release							
	Major software release							

3.1.1 IE Identifier

Transparent Data. See definitions at section 2.5.

3.1.2 Car Manufacturer ID

Identifies the vehicle in which the hardware/software product is resident. This informative to the receiver of the message who should act upon this information if it is a version it cannot handle.

Value	Description	Value	Description
0	Mercedes-Benz	36	Suzuki Veículos do Brasil
1	Renault	37	Troller
2	Chevrolet	38	TAC
3	Porsche	39	Smart
4	Ford	40	Ssangyong
5	PSA	41	SEAT
6	Jaguar	42	Agrale
7	Fiat	43	Chana
8	Volkswagen	44	Mazda
9	Mitsubishi	45	Mahindra
10	Honda	46	Daewoo
11	Toyota / Lexus	47	Mini
12	Nissan	48	Daihatsu
13	Ford Trucks	49	Aston Martin
14	Volkswagen Trucks	50	Ferrari
15	Yamaha Motorcycles	51	Bugati
16	Scania	52	Lamborghini
17	Volvo Brasil	53	Maseratti
18	Iveco	54	MV Agusta Motorcycles
19	Hyundai	55	Piaggio Motorcycles
20	Kia Motors	56	Triumph Motorcycles
21	Subaru	57	Husqvarna Motorcycles
22	Chrysler	58	FYM Motorcycles
23	BMW	59	Ducati Motorcycles

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24	Audi	60	Buell Motorcycles
25	Harley-Davidson	61	Aprila Motorcycles
26	Honda Motorcycles	62	Skoda
27	Suzuki Motorcycles	63	Bentley
28	Dafra	64	Others
29	Kasinski	65	Bramont
30	Sundown Motorcycles	66	International
31	Traxx	67 - 98	Not currently used
32	Effa Motors	99	All
33	Volvo Car	100-126	Not currently used
34	Land Rover	127	Not Specified (SIMRAV ID)
35	Alfa Romeo		

Table 3 – Version Element: Car Manufacturer IDs.

3.1.3 TCU Manufacturer ID

This just identifies the source of the hardware/software product. It is informational. This informative to the receiver of the message who should act upon this information if it is a version it can not handle.

Value	Definition
0	Reserved
1	Motorola
2	Nokia
3	Becker
4	Clarion
5-127	Not currently used
128	Continental
129	Delphi
130	Johnson Controls
131	Kostal
132	Magneti Marelli
133	PST
134	Visteon
135	Quanta
136	Compsis
137	Maxtrack
138	Trackware
139	Cellocator
140	Aritronix
141	Ituran
142	Others
143-255	Reserved

Table 4 – Version Element: TCU Manufacturer IDs.

3.1.4 Major Hardware Release

This is a binary number that identifies the current revision of the hardware. This informative to the receiver of the message who should act upon this information if it is a version it cannot handle.

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3.1.5 Major Software Release

This is a binary number that identifies the current revision of the software release. This informative to the receiver of the message who should act upon this information if it is a version it cannot handle.

Example:

Octet / Bit	0	1	2	3	4	5	6	7
Version Element	IE Identifier = 0		More Flag = 0	Length = 4				
	Car Manufacturer ID = 8							
	TCU Manufacturer ID = 131							
	Major hardware release = 1							
	Major software release= 3							

5 octet message binary data – Car manufacturer = Volkswagen, TCU manufacturer = Kostal, Hardware release = 1 and Software release = 3 → 00000100 00001000 1000011 00000001 00000011

3.2 Timestamp Element

This element specifies the date and time information.

Structure:

Octet \ bit	0	1	2	3	4	5	6	7
1	Year(00=1990, 01=1991 ranges up to 62=2052)						Month of year range (1...12)	
2	Month of year		Day of month, range (1...31)					Hour of day
3	Hour of day, range (0...23)				Minutes, range (0...59)			
4	Minutes		Seconds, range (0...59)					

3.2.1 Year

This specifies the year beginning in 1990. The field allows for a year of up to 2052. The value of 63 is reserved for future expansion if needed. (6 bits)

3.2.2 Month

This field represents the month of the year which ranges from 1 to 12 where 1 is January and 12 is December. (4 bits)

3.2.3 Day

This field represents the day of the month and ranges from 1 to 31. (5 bits)

3.2.4 Hour

This field represents the hour of the day and ranges 0 to 23. (5 bits)

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3.2.5 Minute

This field represents the minute of the hour and ranges from 0 to 59. (6 bits)

3.2.6 Seconds

This field represents the seconds of the minute and ranges from 0 to 59.(6 bits)

Example: March 14th, 2008 – 3:48:56 pm

Year = 2008 → 18 → 010010 (6 bits)

Month = March → 3 → 0011 (4 bits)

Day = 14 → 14 → 01110 (5 bits)

Hour = 15 → 15 → 01111 (5 bits)

Minute = 48 → 48 → 110000 (6 bits)

Second = 56 → 56 → 111000 (6 bits)

Message: 01001000 11011100 11111100 00111000

3.3 TCU Descriptor Element

Structure:

Octet \ bit	0	1	2	3	4	5	6	7
TCU Descriptor Element	IE Identifier		More Flag	Length				
	Reserved							
	IE Identifier		More Flag	Length				
	Device ID							
	IE Identifier		More Flag	Length				
	Version ID							

3.3.1 Device ID

This describes the device for which we want the version number.

Value	Definition
1	TCU hardware version number
2	TCU hardware manufacturer number
3	TCU software version number
4	TCU CAN version number
5	ACP transport layer version number
6	ACP application layer version number
7-255	Reserved

Table 5 - TCU Descriptor Element: Device IDs

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3.3.2 Version ID

This specifies the version information for the specified device. The version number can be a number and taken from the following version table, or it can be a character string for version not in the table. The currently defined decimal versions are

Value	Version
1	1
2	2
3	3
4	4
5	5
:	:
n-255	

3.4 Vehicle Descriptor Element

The intent of this information element is to define the vehicle to the service operator.

Structure:

Octet \ bit	0	1	2	3	4	5	6	7
1	IE Identifier = 0		More flag	Length				
2	Vehicle Descriptor Flag1							
3	Vehicle Descriptor Flag2							
4	Language							
5	Model year							
6..p-1	VIN							
p..n-1	TCU serial #							
n..m-1	License Plate							
m..k-1	Vehicle Color							
k..j-1	Vehicle Model							
J...l-1	IMEI							
l...h-1	SIM Card ID (ICCID)							
H...i-1	Auth. Key							

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Octet \ bit	0	1	2	3	4	5	6	7
Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

3.4.1 Flags

In the above table, a bit value of 0 indicates the element is absent while a bit value of 1 indicates the element is present. If the bit indicates the element is present, but the length of its element is 0, then the field has no data following it. If the length is not 0, then the element specifies the new value. If all bits are 0, then no elements are present, and only 1 octet of storage is used.

Field Names	Description
VIN	Vehicle Number Identification (optional)
TCU Serial	Serial number (optional)
IMEI	IMEI identification number (optional)
SIM Card	SIM Card number - ICCID (mandatory)
Vehicle Color	Vehicle color (optional)
Vehicle Model	Vehicle model (optional)
License Plate	License plate (optional)
Language	Language (optional)
Model Year	Model year (optional)
Auth. Key	Auth.Key (mandatory only on Telematic Service Activation/Deactivation message)

Table 6 - Vehicle Descriptor Element: Fields

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3.5 Error Element

This element represents error information.

Structure:

Octet \ bit	0	1	2	3	4	5	6	7
Error Element	IE Identifier = 0		More Flag	Length				
	Error Code							

3.5.1 Valid Error Codes

Code	Description
0	Everything OK, no mistake
1	Service currently not available
2	Incorrect application
3	Unknown version
4	Unknown message type
5	Unknown data in message
6	Unknown transport version
7	Data error in transport frame
8	Security violation
9	No access; no customer
10	No access; service not available
11	Not access; authentication failure
12	No access; other reasons
13	Invalid session ID
14	Reserved
15	Language not supported
16	Descriptor in <i>Provision Update Message</i> does not match data in the TCU
17	SIM identifier in the TCU is not the provisioned SIM
18	Provisioning layer received, but unable to process
19	General non-specific error
20	No access to protected phonebook on SIM card
21	Unable to write to EEPROM
22	Invalid phone number
23	VIN does not match
24	Vehicle type does not mat
25	Provisioning request not processed. Too many target applications specified. Provision with a smaller group.
26	Missing phone number
27	Invalid TCU Service Activation

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28	Invalid TCU Service Deactivation
29	Buffer Overflow
30...255	Available

Table 7 - Error Element: Error Codes.

3.6 Control Function Element

This element represents the remote vehicle entity to control.

Structure

Octet \ bit	0	1	2	3	4	5	6	7
Control Function	IE Identifier=0		More Flag	Length				
	Entity ID							
	Reserved				Transmit Units			
	Transmit Interval							

3.6.1 Entity ID

Controlled Entity

ID	Description
0	Door Locks
1	Vehicle Tracking
2	Covert Mode
3	Microphone
4	Reserved
5	Transmit Interval
6	Reserved
7	Vehicle Tracking with commit
8	Commit to Vehicle Tracking
9	Vehicle anti theft alarm facilities (lights, horn)
10	Immobilize vehicle (*)
11	Remote door locks function
12	Primary antenna
13	Call Service Operator for provisioning using pre-programmed phone numbers
14	Call Service Operator on data bearer.
15	Fuel Pump Blocking
16	Siren
17	Vehicle position history (#)
18...127	Reserved
128	Anti-theft Blocking (ACP245)
129...255	Reserved

Table 8 - Control Function Element: Entities IDs

(*) Auto immobilizer device

(#) Only used if TCU doesn't send the vehicle position history automatically after receiving a "Configuration TCU Service Activation Message ACP 245" (when the SO is contracted). Further details see Appendix III.

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3.6.2 Transmit Units

It specifies the unit of interval at which a message is sent on a regular period.

Value	Command Definition
0	Second
1	Minute
2	Hour
3	Send message one more time
4	Send only one message

3.6.3 Transmit Interval

It specifies the interval at which a message is sent to SO.

3.7 Function Command Element

This element represents the action required or report the status of the tracking device (tracking entity).

Structure:

Octet \ bit	0	1	2	3	4	5	6	7
Function Command	IE Identifier=0		More Flag	Length				
	Function Command or Status							
	Raw Data							

3.7.1 Function Command or Status

Value	Command Definition	Status Definition
0	Permit	Permitted
1	Reject	Rejected
2	Enable (Start)	Enabled (Started)
3	Disable (Stop)	Disabled (Stopped)
4 (*)	Request	Completed

Table 9 - Function Command Element: Available actions or status.

(*) Request means ask for status and is optional and dependent of the TCU manufacturer implementation.

3.7.2 Raw Data

It is used to transmit any raw data between the TCU and SO.

Octet \ bit	0	1	2	3	4	5	6	7
	IE Identifier=0		More Flag	Length				
	Raw Data							

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3.8 Location Element

This group defines the position history of the vehicle.

(*) If data type 0x0047 = false then Location Element has length = "0"

Octet \ bit	0	1	2	3	4	5	6	7
Location	IE Identifier=0		More Flag	Length				
	Current GPSRawData (3.8.1)							
	Prior GPSRawData (3.8.1)							
	Current Dead Reckoning Data (3.8.16)							
	Array of Area Location Delta Coding (3.8.17)							

The current GPSRawData and prior GPSRawData are only needed when the GPS is co-located with the TCU and, therefore, is optional.

3.8.1 GPSRawData Element

The GPS raw data location. If no data available the field length must be set to 0 and follow octet is next element.

Octet \ bit	0	1	2	3	4	5	6	7
GPSRawData Element	IE Identifier=0		More Flag	Length				
	Area Location Coding (3.8.2)							
	Number of Satellites (0 .. n)				Reserved			
	Satellite ID (0)							
	Satellite ID (n)							

3.8.2 Area Location Coding

Octet \ bit	0	1	2	3	4	5	6	7
Area Location Coding	IE Identifier=0		More Flag	Length				
	More Flag	AreaLocationStatusFlag1						
	More Flag	AreaLocationStatusFlag2						
	Area Type			LocationTypeCoding			Reserved	
	More Flag	Time Difference						
	Longitude (byte 0)							
	Longitude (byte 1)							
	Longitude (byte 2)							
	Longitude (byte 3)							
	Latitude (byte 0)							
	Latitude (byte 1)							
	Latitude (byte 2)							
	Latitude (byte 3)							
	Altitude (byte0)							

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	Altitude (byte 1)		
	Position Uncertainty Estimate		K/ HDOP
	Heading Uncertainty Estimate	Heading	
	Reserved set to 0	Distance Flag	Time Flag
	Velocity		

3.8.3 AreaLocationStatusFlag1

Bit	Definition
1	Reserved and set to 0
2	1=No 3D fix available 0=Is using 3D fix
3	1=No 2D fix available 0=Is using 2D fix
4	1=Position data is not valid 0=Position data is valid
5	1=Differential GPS is being used 0=Differential GPS not being used
6	1=Heading data is not valid 0=Heading data is valid
7	1=Almanac is bad 0=Almanac is good

3.8.4 Area Location Status Flag 2

Bit	Definition
1	Reserved, set to 0
2	0=old GPS data used from satellites 1=new GPS data from satellites
3	Reserved, set to 0
4	Reserved, set to 0
5..7	0=North 1=North East 2=East 3=South East 4=South 5=South West 6=West 7=North West

3.8.5 Area Type

Area Type Definitions for Location and Deltas.

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Value	Definition
0	Point (latitude, longitude, altitude) in milliarcseconds increments
1	Point (latitude, longitude, altitude) in 100 milliarcseconds increments
2-7	Reserved

3.8.6 LocationTypeCoding

The element *location type coding* defines the coordinate system used to define the location of the TCU.

Value	Definition
0	WGS 84
1-7	Reserved

3.8.7 Time Difference

The time difference is the time that has elapsed between a prior GPS reading and the current GPS reading or the time difference between the current GPS reading and the actual cause for transmission. The default value is in seconds unless defined by Time Flag. The value = 0 means that this field is NULL or not calculated.

3.8.8 Longitude

The longitude may vary from -180° to +180°. The number 0° corresponds to the Greenwich meridian. The value may range from -648,000,000 to +648,000,000. The units are in millisecondseconds and allow for the range from -648,000,000 to +648,000,000. One unit corresponds to approximately .03 meter. Positive values are East of the Greenwich median and negative values are West. Negative numbers are expressed in 2's complement.

3.8.9 Latitude

The latitude may vary from -90° to +90°. The number corresponds to the equator. The units are in milliarcseconds and allow for the range from -324,000,000 to +324,000,000. One unit corresponds to approximately .03 meter. Positive values are North and negative values are South. Negative numbers are expressed in 2's complement.

3.8.10 Altitude

The altitude may vary from -1000m to 18,000 meters. The number 0 corresponds to sea level. One unit corresponds to 1 meter.

3.8.11 Position Uncertainty Estimate

In the Position Uncertainty Estimate field there are 7 bits for data and 1 bit to select between K or HDOP:

bit value (K/HDOP)	Description
0	K
1	HDOP

3.8.11.1 Position Uncertainty Estimate using K value

The *position uncertainty estimate using K value* is defined by a variation on the Binomial expansion. The uncertainty, expressed in meters, is mapped to a number K, with the following formula $R = C((1+x)^K - 1)$.

Value of K	Uncertainty
0	0 m
1	1 m
2	2,1 m
...	

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20	57,3 m
...	
40	443 m
...	
60	3 km
...	
127	1800 km

3.8.11.2 Position Uncertainty Estimate using HDOP value

The *position uncertainty estimate using HDOP value* is defined according table below that is calculated by GPS module.

HDOP Value	Rating	Description
1	Ideal	This is the highest possible confidence level to be used for applications demanding the highest possible precision at all times
2-3	Excellent	At this confidence level, positional measurements are considered accurate enough to meet all but the most sensitive applications
4-6	Good	Represents a level that marks the minimum appropriate for making business decisions. Positional measurements could be used to make reliable in-route navigation suggestions to the user
7-8	Moderate	Positional measurements could be used for calculations, but the fix quality could still be improved. A more open view of the sky is recommended
9-20	Fair	Represents a low confidence level. Positional measurements should be discarded or used only to indicate a very rough estimate of the current location
21-50	Poor	At this level, measurements are inaccurate by as much as 300 metres with a 6 meter accurate device (50 dop * 6 meters) and should be discarded
>50-127	No Signal/ Discard	At this level, measurements are inaccurate-discard

3.8.12 Heading Uncertainty Estimate

The *heading* is defined as the azimuth that means the angle between the horizontal plan and an element. The *heading uncertainty estimate* is defined in the following table.

Value	Heading Uncertainty
0	$\leq 6^\circ$
1	$\leq 9^\circ$
2	$\leq 12^\circ$
3	$\leq 15^\circ$
4	$\leq 20^\circ$
5	$\leq 25^\circ$
6	$\leq 30^\circ$
7	$> 30^\circ$ or NULL

3.8.13 Heading

The heading has multiples of 15 degrees. For example, a heading value of 3 is 45 degrees or North East.

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3.8.14 Distance Flag

Value	Definition
0	Units are not defined
1	Kilometers
2	Miles

3.8.15 Time Flag

Value	Definition
0	Seconds
1	Minutes
2	Hours
3	Reserved

3.8.16 Current Dead Reckoning Data

The Dead Reckoning Data follows the WGS4 format. The specific format for Dead Reckoning is extracted from the WGS84 reference ellipsoid. It is the current position including the dead reckoning data. If no data available the field length must be set to 0 and follow octet is next element.

Octet \ bit	0	1	2	3	4	5	6	7
Current Dead Reckoning Data	IE Identifier=0		More Flag	Length				
	Latitude							
	Longitude							

3.8.17 Array of Area Location Delta Coding

This element defines a more compact form for specifying the position history of a vehicle. If no data available the field length must be set to 0 and follow octet is next element.

Octet \ bit	0	1	2	3	4	5	6	7
Area Location Delta Coding	IE Identifier=0		More Flag	Length				
	More Flag		Delta Longitude 1					
	More Flag		Delta Latitude 1					
	...		...					
	More Flag		Delta Longitude n					
	More Flag		Delta Latitude n					

This 1 octet fields allow for a value range of –64 to +63 milliarcseconds or 100 milliarcseconds increments as defined by Area Type in section 17.16. This provides for 3.1 meter resolution which is an acceptable compromise for standard maps having 5 meter resolution. This also reduces the computational requirements of the TCU. Since the range for milliarcseconds is ± 63 (100s) milliarcseconds, this equates to a distance of ± 195 meters at the equator and gets smaller as the north/south pole is approached.

The deltas are always paired as a Longitude and Latitude. The first delta location (Delta 1) is relative to the Prior GPS Raw Data location. The second delta location is relative to the first delta location. Delta N location is first location recorded and is relative to Delta N-1. Positive deltas indicate a direction in the north and east directions.

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3.9 Breakdown Status Element

This group defines the nature of the breakdown.

Structure:

Octet \ bit	0	1	2	3	4	5	6	7
Breakdown Status Element	IE Identifier=0		More Flag	Length				
	More Flag=1	Breakdown Source (<i>First flag</i>)						
	More Flag=1	Breakdown Source (<i>Second flag</i>)						
	More Flag (*)	Breakdown Source (<i>Third flag</i>)						
	More Flag	Breakdown Sensor						
	IE Identifier		More Flag	Length				
	Breakdown Data							

(*) More Flag default = 0 – Defines the use of first, second and third flag only. More flag = 1 means that there is additional breakdown sources flags defined by TCU suppliers.

3.9.1 Breakdown Source

This octets flags fields define the cause for the breakdown. Breakdown Source is activated only when an Event or Alarm occurs.

First Flag

Bit	Breakdown Definition
0	Additional flag
1	Manually activated
2	Vehicle rolled
3	Air bag activated
4	Crash sensor activated
5	Floating car data input
6	Tow truck needed
7	Vehicle initiated theft tracking (Theft alarm triggered)

Second Flag

Bit	Breakdown Definition
0	Additional flag
1	Vehicle is started (ignition on)
2	Vehicle is turned off (ignition off)
3	Vehicle is moved
4	Other sensor activated
5	Re-send location (TCU button pressed)
6	Re-send location (SO sent message)
7	Unauthorized vehicle movement as defined by Theft Alarm

Third Flag

Bit	Breakdown Definition
0	Additional flag
1	1=Siren off

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2	1=Siren on
3	1=Main battery is reconnected
4	1=Main battery is disconnected
5	0=Panic button is off 1=Panic button is on
6	1=Blocking is on
7	1=Blocking is off

3.9.2 Breakdown Sensor

For automatically detected emergencies, this byte defines the sensor within the class *Breakdown source* that initiated the emergency request.

Bit	Crash sensors definition
0	Additional flag
1	Rollover sensor activated
2	Front sensor
3	Rear sensor
4	Side sensor
5	Vehicle Alarm activated
6	Reserved
7	Breakdown data = 1 (Status)

3.9.3 Breakdown Data

When Breakdown sensor bit 7 is set to 1, the format of breakdown data field is the same as described in item 3.9.1 and represents breakdown source status.

3.10 Information Type Element

Structure:

Octet \ bit	0	1	2	3	4	5	6	7
Information Type Element	IE Identifier=0		More Flag	Length				
	Addl Flag	Information Type						
	IE Identifier		More Flag	Length				
	Raw Data							

3.10.1 Information Type

This flag defines the type of proprietary information packed as raw data.

Value	Definition
0	Reserved
1	Verbal Information
2	Stock Information
3	Travel Route Information
4	Hotel Information
5	Traffic information requested (verbal)
6	Traffic information requested (automated)
7	ASCII Text String

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8	Point of Interest
9	Cargo
10	Private
11	Environmental Data
12	Timestamp
13	Mobile country code
14	Menu button
15..127	Reserved

Table 10 - Information Type Element: Valid information types.

3.10.2 Raw Data

Raw data transmitted in information type element.

3.11 TCU Data Element

Structure:

Octet \ bit	0	1	2	3	4	5	6	7
TCU Data Element	IE Identifier =0		More Flag	Length				
	Data Type MSB (0)							
	Data Type LSB (0)							
	Length Data Type (0)							
	Configuration Data (0)							
	...							
	Data Type MSB (n)							
	Data Type LSB (n)							
	Length Data Type (n)							
	Configuration Data (n)							

3.11.1 Data Type

This element describes the configuration parameter index to be changed. See definitions at appendix I (Available Configuration Parameters Indexes).

3.11.2 Length Data Type

This element means the length of the parameters data.

3.11.3 Configuration Data

This defines the new configuration parameter value to be changed.

3.12 TCU Data Error Element

Structure:

Octet \ bit	0	1	2	3	4	5	6	7
TCU Data Error Element	IE Identifier =0		More Flag	Length				
	Data Type MSB (0)							
	Data Type LSB (0)							
	Length Data Type (0)							
	Configuration Data (0)							

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	Error Element (0)
	...
	Data Type MSB (n)
	Data Type LSB (n)
	Length Data Type (n)
	Configuration Data (n)
	Error Element (n)

3.12.1 Data Type

This element describes the configuration parameter index to be changed. See definitions at appendix I (Available Configuration Parameters Indexes).

3.12.2 Length Data Type

This element means the length of the parameters data.

3.12.3 Configuration Data

This defines the new configuration parameter value to be changed.

3.12.4 Error Element

This field is previously defined in item 3.5.

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4 HEADER DESCRIPTION

Octet / Bit	0	1	2	3	4	5	6	7
1	Reserved Set to 0	Private flag	Application ID					
2	Reserved Set to 0	Private flag	Test Flag	Message Type				
3	Version Flag	Version			Message Control Flag			
4 (optional)	More flag	Reserved Set to 0					Message Priority flag	
5, 6	Message Length							

Table 11 - Header Structure (Binary)

4.1 Message Elements Definition

4.1.1 Private flag

ACP Protocol reserved field. The default value is 0 and is used for standard ACP 245 messages. Value = 1 is used for private messages.

4.1.2 Application ID

See section 1.1.

4.1.3 Test Flag

If the value is 0, then this message is release message. If the value is 1, then this message is a test message and its implementation is not finished.

4.1.4 Message Type

This field defines the message type. It is unique for a specific message. If a message is constructed with a unique combination of message elements then that message will have a unique message type. If the *Test Flag* bit is set, then this is a test message.

4.1.5 Version Flag

This is intended to allow for extended control information in the header that is common among many applications. If the value is 0, then this is the last byte of the header before next field data. If the value is 1, then another version control octet of information follows this octet.

4.1.6 Version

This is the version of the application. This version number allows the application to be revised and identified independent of the overall ACP version number. The value follows as identified in the table below:

Value	ACP 245 Version
0	V 1.2
1	V 1.2.1
2	V 1.2.2
3-6	Available
7	Reserved

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In case of the Version Value reach Seven in the table (7 – Reserved), then the representation of the Version Value will continue in the Extended Version Number Field. See example in the Appendix II 13.7.1.6

4.1.7 Message Control Flag

These bit definitions define various actions that are common over many applications.

Bit Number	Definition
0	Reserved, set to 0
1	0 = Use TLV (Type, Length, Value / Variable)
2	0 = The message length field is 8 bits 1 = The message length field is 16 bits
3	0 = Application level response is not expected 1 = Application level response is expected

Message Control Flag			
BIT 0	BIT 1	BIT 2	BIT 3

4.1.8 Message Priority Flag

This flag defines the behavior of sender. This field requires version flag with the value 1.

Value	Definition
0	Reserved
1	Abort: the sender wants to abort the connection. The message length must be 0. Otherwise, the <i>error element</i> follows.
2	Pause: the sender wants to pause the communication.
3	Resume: the sender wants to resume the communication.

4.1.9 Message Length

The message length is the total number of octets in the message including the header.

4.1.10 More Flag

See section 2.3.

4.1.11 Reserved (Used to Extended Version Number of ACP245).

The 1st bit to 5th bit of 4th octet in the Header Element is defined as Reserved. It will be used to represents the Extended Version Number after Version 6th of the Protocol ACP245.

Value	ACP 245 Version
7 - 32	Available

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5 PROVISIONING SERVICE (APPLICATION ID = 1)

These are messages that allow the service operator SO and the TCU in the vehicle to provision applications. This allows for the establishment of features and services by application. Parameter updates are usually done within an application. This can include subscription information depending on the business model of the SO. It can cover the means whereby the TCU customer pays (e.g. on a usage basis or for a fixed period basis or some other combination). If the service operator wants to sign up a customer when the service is 1st used, then more information must be supplied as part of the application, and information the customer may not readily have at time of use. [1]

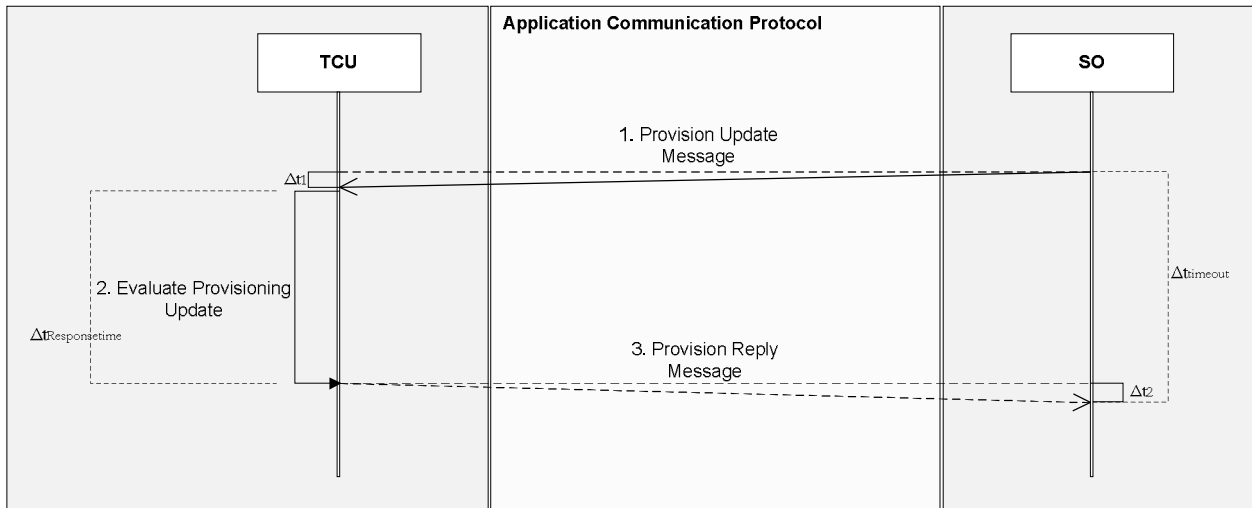


Figure 1- Provisioning Application Message Flow.

5.1 Provisioning Message Set

Message Type Id	Message Type
01	Provision Update Message #1
02	Optional Provision Update Commit Message (*)
03	Provision Reply Message #1
04	Optional Provision Reply Commit Message #2 (*)
05	Provision Request Message (*)
06	Provision Status Message (*)
07	Provision Update Message #2 (*)
08	Provision Reply Message # 2 (*)

Remarks:

1. (*)Only for ACP 3.1.0.2.
2. The DENATRAN Decret n°129/2008 requires as mandatory the ACP Applications 1, 2, 6, 10 and 11. It is mandatory to TCU manufacturer to implement the above mentioned applications and there is no need to the SO to “provision” them. Any attempt to deactivate them must be rejected.
3. Message types 01 and 03 are required for ACP 245.

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5.2 Provision Update Message #1 (From SO to TCU)

General Message Format:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION	Header Element							
	Version Element							
	More Flag	Target Application ID						
	ApplFlag1		ControlFlag1					
	ControlFlag2							
	Start Time (Timestamp Element)							
	End Time (Timestamp Element)							
	Grace Time (Timestamp Element)							
	TCU Descriptor Element							
	Vehicle Descriptor Element							

Table 12 - Provision Update Message: elements.

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard	Application ID (default=0x01)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type (default=0x01)				
	Version Flag	Application Version			Message Control Flag (bit 3=1)			
	Message Length							

Version Element	IE Identifier =0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Message Fields	More Flag	Target Application ID									
	ApplFlag1				ControlFlag1						
	ControlFlag2										

Start Time (Timestamp)	Year (00=1990, 01=1991 ranges up to 62=2052)				Month of year range (1...12)	
	Month of year		Day of month, range (1...31)			Hour of day
	Hour of day, range (0...23)			Minutes, range (0...59)		
	Minutes		Seconds, range (0...59)			

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End Time (Timestamp)	Year (00=1990, 01=1991 ranges up to 62=2052)		Month of year range (1...12)	
	Month of year	Day of month, range (1...31)		Hour of day
	Hour of day, range (0...23)		Minutes, range (0...59)	
	Minutes	Seconds, range (0...59)		

TCU Descriptor Element	IE Identifier	More Flag	Length
	Reserved		
	IE Identifier	More Flag	Length
	Device ID		
	IE Identifier	More Flag	Length
	Version ID		

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

5.2.1 Message Elements Definition

5.2.1.1 Header Element

See definitions at section 4.1. In Message Control Flag, bit 3 is always set to “1”.

5.2.1.2 Version Element

See definitions at section 3.1.

5.2.1.3 Message Fields

5.2.1.3.1 Target Application ID

It is used to identify the application that is to be provisioned. The Additional Flag is used to provide a list of Application Ids to be provisioned with the same data. The Provisioning and Configuring target Applications Ids should not be used in the Provision update message. In Application=2, Message Type=8 it must be used Target Application ID=0.

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5.2.1.3.2 ApplFlag1

This table defines the provisioning to perform.

Value	Definition
0	No change to application
1	Activate application
2	Deactivate application
3	Change for this application

The value 0 (zero) is used to realize a consult or a request about the currently configuration status of a specific target application identified by the field target application id.

The value 1 (one) will activate (provisioning) the target application, while the value 2 (two) will deactivate (not provisioning) the currently target application identified by the field target application id.

The 3 (three) value will request a change or update of the target application pointed by the field target application id.

5.2.1.3.3 Control Flag 1

This is defined as a 6 bit field with bit 0 on the left and bit 5 on the right. Remember that this field does not begin on a byte boundary.

Bit Number	Definition
0	1=A 2nd byte ControlFlag2 exists 0=This is the last Control Flag
1	1=Grace Time is present 0=Grace Time is absent
2	1=Start Time field is present 0=Start Time field is absent
3	1=End Time or Expiration Time field is present 0=End Time or Expiration Time field is absent
4	1=Vehicle Descriptor element is present 0=Vehicle Descriptor element is absent
5	1=Provisioning uses <i>provision update commit message</i> 0=Provisioning does not use <i>provision update commit message</i>

Control Flag 1					
BIT 0	BIT 1	BIT 2	BIT 3	BIT 4	BIT 5

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5.2.1.3.4 Control Flag 2

Bit Number	Definition
0	Additional Flag
1..3	0=Use the profile method of collection 1=Use the sample method of collection 2..7=Reserved
4	0=Number of samples is not present 1=Number of samples is present
5..7	0=No sample unit 1=Sample units are in minutes 2=Sample units are in kilometers

Control Flag 2							
BIT 0	BIT 1	BIT 2	BIT 3	BIT 4	BIT 5	BIT 6	BIT 7

5.2.1.4 Start Time (Timestamp Element)

This is the time at which service is to start. If it is absent, then it is the current time or another predefined time. See definitions at section 3.2.

5.2.1.5 End Time (Timestamp Element)

This is the time at which service is to end. If it is absent, then it no end time is defined. This field is only required when activating a service and establishing the ending time. If a service is to be deactivated and this element is not present, then the deactivation is immediate. If this element is present for a service deactivation, then it specifies the date where the service ends. See definitions at section 3.2.

5.2.1.6 Grace Time (Timestamp Element)

If supported in the TCU, this is the time after end time whereby service for the application is still supported. See definitions at section 3.2

5.2.1.7 TCU Descriptor Element

See definitions at section 3.3.

5.2.1.8 Vehicle Descriptor Element

See definitions at section 3.4.

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5.3 Provision Reply Message #1 (From TCU to SO)

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION Message fields	Header Element							
	Version Element							
	More Flag	Target Application ID						
	ApplFlag1		ControlFlag1					
	Status Flag1		TCU Response Flag			Reserved 0)		
	Error Element							
	Vehicle Descriptor Element							

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard Flag	Application ID (default=0x01)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type (default=0x03)				
	Version Flag	Application Version			Message Control Flag			
	Message Length							

Version Element	IE Identifier =0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Message Fields	More Flag	Target Application ID						
	ApplFlag1		ControlFlag1 (default=2)					
	Status Flag1		TCU Response Flag			Reserved (default=0)		

Error Element	IE Identifier =0	More Flag	Length
	Error Code		

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Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

5.3.1 Message Elements Definition

5.3.1.1 Header Element

See definitions at section 4.1.

5.3.1.2 Version Element

See definitions at section 3.1.

5.3.1.3 Message Fields Element

Initial definitions are at section 5.2.1.3.

5.3.1.3.1 StatusFlag1

This indicates the current provisioning status of the application.

Value	Definition
0	Application already provisioned
1	Application not already provisioned
2	See Error element for more details at 3.5
3	Reserved

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5.3.1.3.2 TCU Response Flag

This indicates the reason for this reply message.

Value	Definition
0	Reserved
1	TCU initialize mode
2	TCU response to <i>provision update message</i>
3	TCU response to <i>provision commit message</i>

5.3.1.4 Error Element

See definitions at section 3.5.

5.3.1.5 Vehicle Descriptor Element

See definitions at section 3.4.

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6 CONFIGURATION (APPLICATION ID = 2)

These are messages that allow the service operator (SO) and the TCU in the vehicle to configure applications. This allows for the establishment of features and services by application. [1]

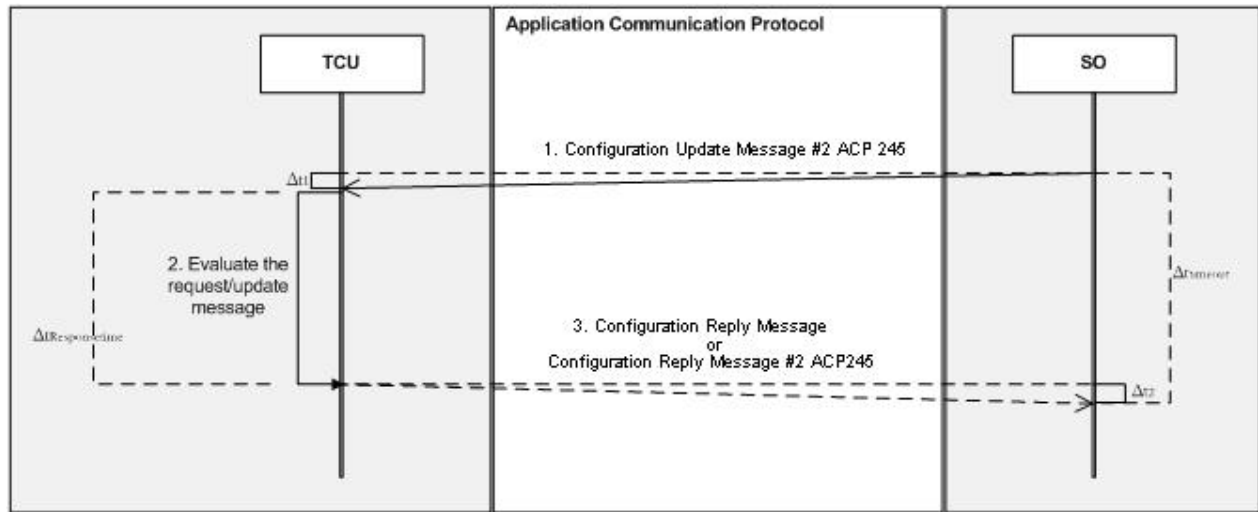


Figure 2 - Configuration Message Flow.

6.1 Configuration Message Set

Message Type Id	Message Type
01	Configuration Update Message (*)
02	Optional Configuration Update Commit Message (*)
03	Configuration Reply Message
04	Optional Configuration Reply Commit Message (*)
05	Configuration Request Message (*)
06	Configuration Status Message (*)
07	Configuration Edit Message (*)
08	Configuration Update Message #2 ACP 245
09	Configuration Reply Message #2 ACP 245
10	Configuration TCU Service Activation/Deactivation Message ACP 245

Remarks:

1. (*)Only for ACP 3.1.0.2.
2. Message types 03, 08, 09 and 10 are required for ACP 245.

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

6.2 Configuration Update Message #2 ACP 245 (From SO to TCU)

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION <i>Message Fields</i>	Header Element							
	Version Element							
	More Flag	Target Application ID						
	ApplFlag1		ControlFlag1					
	ControlFlag2							
	Reserved							
	Start Time (Timestamp Element)							
	End Time (Timestamp Element)							
	Grace Time (Timestamp Element)							
	Vehicle Descriptor Element							
	TCU Descriptor Element							
	TCU Data Element							

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header Element	Reserved Set to 0	Non Standard Flag	Application ID (default=0x02)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type (default=0x08)				
	Version Flag	Application Version			Message Control Flag (bit 3=1)			
	Message Length							

Version Element	IE Identifier =0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Message Fields	More Flag	Target Application ID = 0									
	ApplFlag1				ControlFlag1						
	ControlFlag2										
	Reserved										

Start Time (Timestamp)	Year (00=1990, 01=1991 ranges up to 62=2052)				Month of year range (1...12)	
	Month of year	Day of month, range (1...31)			Hour of day	
	Hour of day, range (0...23)			Minutes, range (0...59)		
	Minutes	Seconds, range (0...59)				

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End Time (Timestamp)	Year (00=1990, 01=1991 ranges up to 62=2052)		Month of year range (1...12)	
	Month of year	Day of month, range (1...31)		Hour of day
	Hour of day, range (0...23)		Minutes, range (0...59)	
	Minutes	Seconds, range (0...59)		

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIM Card ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

TCU Descriptor Element	IE Identifier	More Flag	Length
	Reserved		
	IE Identifier	More Flag	Length
	Device ID		
	IE Identifier	More Flag	Length
	Version ID		

TCU Data Element	IE Identifier	More Flag	Length
	Data Type MSB (0)		
	Data Type LSB (0)		
	Length Data Type (0)		
	Configuration Data (0)		
	...		
	Data Type MSB (n)		
	Data Type LSB (n)		
	Length Data Type (n)		
	Configuration Data (n)		

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6.2.1 Message Elements Definition

6.2.1.1 Header Element

See definitions at section 4.1. In Message Control Flag, bit 3 is always set to “1”.

6.2.1.2 Version Element

See definitions at section 3.1.

6.2.1.3 Message Fields Element

See definitions at section 5.2.1.3. Target Application ID is always set to “0”.

6.2.1.4 Start Time (Timestamp Element)

This is the time at which service is to start. If it is absent, then it is the current time or another predefined time. See definitions at section 3.2.

6.2.1.5 End Time (Timestamp Element)

This is the time at which service is to end. If it is absent, then no end time is defined. This field is only required when activating a service and establishing the ending time. If a service is to be deactivated and this element is not present, then the deactivation is immediate. If this element is present for a service deactivation, then it specifies the date where the service ends. See definitions at section 3.2.

6.2.1.6 Grace Time (Timestamp Element)

If supported in the TCU, this is the time after end time whereby service for the application is still supported. See definitions at section 3.2.

6.2.1.7 Vehicle Descriptor Element

See definitions at section 3.4.

6.2.1.8 TCU Descriptor Element

See definitions at section 3.3.

6.2.1.9 TCU Data Element

See definitions at section 3.11.

6.3 Configuration Reply Message (From TCU to SO)

Used for a single error element response.

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION Message Fields	Header Element							
	Version Element							
	Reserved							
	More Flag	Target Application ID						
	ApplFlag1		ControlFlag1 (default=2)					
	Status Flag1		TCU Response Flag			Reserved		
	Error Element							
	Vehicle Descriptor Element							

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Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard Flag	Application ID (default=0x02)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type (default=0x03)				
	Version Flag	Application Version			Message Control Flag			
	Message Length							

Version Element	IE Identifier =0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Reserved								
----------	--	--	--	--	--	--	--	--

Message Fields	More Flag	Target Application ID						
	ApplFlag1		ControlFlag1 (default=2)					
	Status Flag1		TCU Response Flag			Reserved		

Error Element	IE Identifier =0	More Flag	Length
	Error Code		

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

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6.3.1 Message Elements Definition

6.3.1.1 Header Element

See definitions at section 4.1.

6.3.1.2 Version Element

See definitions at section 3.1.

6.3.1.3 Message Fields Element

See definitions at section 5.2.1.3.

6.3.1.4 Error Element

See definitions at section 3.5.

6.3.1.5 Vehicle Descriptor Element

See definitions at section 3.4.

6.4 Configuration Reply Message #2 ACP 245 (From TCU to SO)

Used for a multiple error element response.

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION Message Fields	Header Element							
	Version Element							
	More Flag	Target Application ID						
	ApplFlag1		ControlFlag1 (default=2)					
	Status Flag1		TCU Response Flag			Reserved		
	TCU Data Error Element							
	Vehicle Descriptor Element							

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard Flag	Application ID (default=0x02)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type (default=0x09)				
	Version Flag	Application Version			Message Control Flag			
	Message Length							

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Version Element	IE Identifier =0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Message Fields	More Flag	Target Application ID	
	ApplFlag1	ControlFlag1 (default=2)	
	Status Flag1	TCU Response Flag	Reserved

TCU Data Error Element	IE Identifier	More Flag	Length
	Data Type MSB (0)		
	Data Type LSB (0)		
	Length Data Type (0)		
	Configuration Data (0)		
	Error Element (0)		
	...		
	Data Type MSB (n)		
	Data Type LSB (n)		
	Length Data Type (n)		
	Configuration Data (n)		
	Error Element (n)		

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

6.4.1 Message Elements Definition

6.4.1.1 Header Element

See definitions at section 4.1.

6.4.1.2 Version Element

See definitions at section 3.1.

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6.4.1.3 Message Fields Element

See definitions at section 5.2.1.3.

6.4.1.4 TCU Data Error Element

See definitions at section 3.12.

6.4.1.5 Vehicle Descriptor Element

See definitions at section 3.4.

6.5 Configuration TCU Service Activation/ Deactivation Message ACP 245 (From SO to TCU)

Used to TCU Service Activation and Deactivation. This process is better described at section 10.

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION Message Fields	Header Element							
	APN Configuration Element							
	Server Configuration Element							
	Control Byte1							
	Vehicle Descriptor Element (ICCID and Auth.Key)							

6.5.1 Message Elements Definition

6.5.1.1 Header Element

See definitions at section 4.1. In Message Control Flag, bit 3 is always set to “1”.

6.5.1.2 APN Configuration Element

Octet \ bit	0	1	2	3	4	5	6	7
1	IE Identifier = 0		More flag	Length				
2..p-1	APN Address Element							
p..q-1	APN Login Element							
q..r-1	APN Password Element							

6.5.1.2.1 APN Address Element

Octet \ bit	0	1	2	3	4	5	6	7
1	IE Identifier = 1		More flag	Length				
2..p-1	APN Address (Text Format)							

6.5.1.2.2 APN Login Element

Octet \ bit	0	1	2	3	4	5	6	7
1	IE Identifier = 1		More flag	Length				
2..p-1	APN Login (Text Format)							

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

6.5.1.2.3 APN Password Element

Octet \ bit	0	1	2	3	4	5	6	7
1	IE Identifier = 1		More flag	Length				
2..p-1	APN Password (Text Format)							

6.5.1.3 Server Configuration Element

Octet \ bit	0	1	2	3	4	5	6	7
1	IE Identifier = 0		More flag	Length = 13				
2..5	First Server IP (4 bytes)							
6..7	First Server Port (2 bytes)							
8..11	Second Server IP (4 bytes)							
12..13	Second Server Port (2 bytes)							
14	Protocol ID							

6.5.1.3.4 Protocol ID

Values	Definition
0	ACP 245
1..31	Reserved
32..255	Defined by TCU Manufacturer

6.5.1.4 Control Byte

Octet \ bit	0	1	2	3	4	5	6	7
1	More flag	Control Value	Reserved					

6.5.1.4.5 Control Value

Bit	Definition
0	More Flag bit, set as 0
1	0= Deactivation (reset all configuration to factory defaults) 1= Activation (*)
2..7	Reserved

(*) This only activates Remote Blocking Function, not the Vehicle Location Function (See data type 0x0047).

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

6.5.1.5 Vehicle Descriptor Element

See definitions at section 3.4.

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

7 REMOTE VEHICLE FUNCTION SERVICE (APPLICATION ID = 6)

The remote vehicle function application allows vehicle functions to be controlled remotely by the SO. These functions may include devices such as door lock motors, window motors, trunk release, or modes such as tracking mode, covert mode, etc. In general, a request is made to the SO by either the TCU or the driver for a specific remote vehicle function to be performed. The SO then initiates the remote vehicle function, assuming a successful authentication. [1]

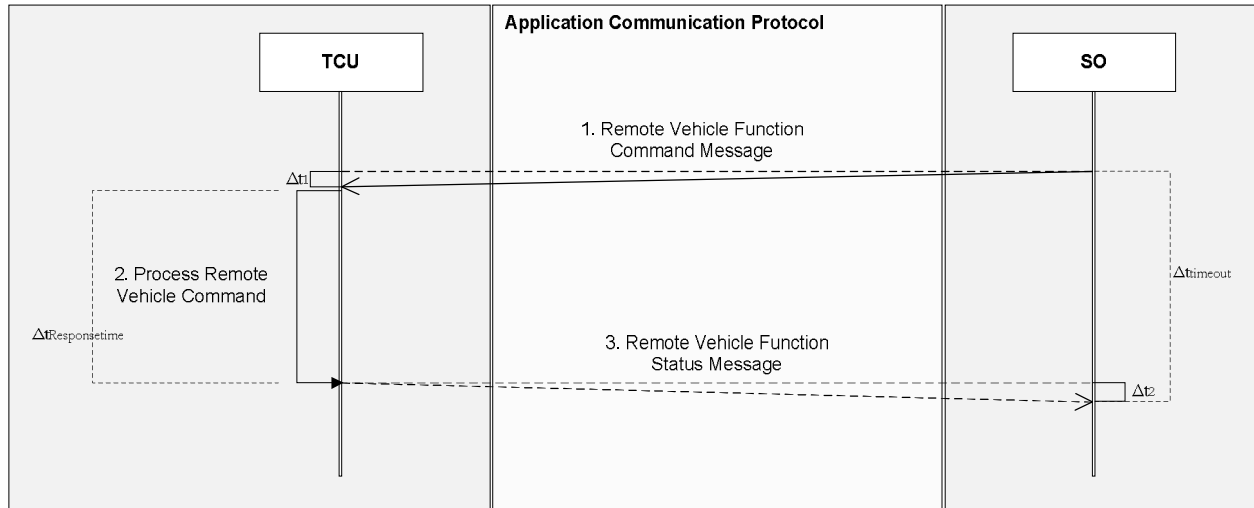


Figure 3 – Remote Vehicle Function Message Flow.

7.1 Remote Vehicle Function Message Set

Message Type Id	Message Type
01	Remote Vehicle Function Request (*)
02	Remote Vehicle Function Command
03	Remote Vehicle Function Status

Remarks:

1. (*)Only for ACP 3.1.0.2.
2. Message types 02 and 03 are required for ACP 245.

7.2 Vehicle Function Command (From SO to TCU)

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION Message Fields	Header							
	Version Element							
	Control Function Element							
	Function Command Element							
	Vehicle Descriptor Element							

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard Flag	Application ID (default=0x06)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type (default=0x02)				
	Version Flag	Application Version			Message Control Flag (bit 3 =1)			
	Message Length							

Version Element	IE Identifier =0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Control Function	IE Identifier	More Flag	Length	
	Entity ID			
	Reserved		Transmit Units	
	Transmit Interval			

Function Status	IE Identifier	More Flag	Length
	Function Command or Status		
	IE Identifier	More Flag	Length
	Raw Data		

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

Example: See appendix II at section **Erro! Fonte de referência não encontrada..**

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

7.2.1 Message Elements Description

7.2.1.1 Header Element

See definitions at section 4.1. In Message Control Flag, bit 3 is always set to “1”.

7.2.1.2 Version Element

See definitions at section 3.1.

7.2.1.3 Control Function Element

See definitions at section 3.6.

7.2.1.4 Function Command Element

See definitions at section 3.7.

7.2.1.5 Vehicle Descriptor Element

See definitions at section 3.4.

7.3 Vehicle Function Status (From TCU to SO)

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION <i>Message Fields</i>	Header							
	Version Element							
	Control Function Element							
	Function Status Element							
	Error Element							
	Vehicle Descriptor Element							

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard Flag	Application ID (default=0x06)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type (default = 0x03)				
	Version Flag	Application Version			Message Control Flag (default=2)			
	Message Length							

Version Element	IE Identifier=0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Control Function	IE Identifier	More Flag	Length	
	Entity ID			
	Reserved		Transmit Units	
	Transmit Interval			

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Function Command	IE Identifier	More Flag	Length
	Function Command or Status		
	IE Identifier	More Flag	Length
	Raw Data		

Error Element	IE Identifier=0	More Flag	Length
	Error Code		

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

Example: See appendix II at section **Erro! Fonte de referência não encontrada..**

7.3.1 Message Elements Description

7.3.1.1 Header Element

See definitions at section 4.1

7.3.1.2 Version Element

See definitions at section 3.1.

7.3.1.3 Control Function Element

See definitions at section 3.6.

7.3.1.4 Function Command Element

See definitions at section 3.7.

7.3.1.5 Vehicle Descriptor Element

See definitions at section 3.4.

7.3.1.6 Error Element

See definitions at section 3.5.

8 VEHICLE TRACKING SERVICE (APPLICATION ID = 10)

The vehicle tracking application sends vehicle location information to the SO on a periodic basis. This enables the SO to track the vehicle in the event that a vehicle was reported missing. Tracking may automatically be enabled in a SO response of another application. However, in the event a vehicle owner reports the vehicle missing, the SO must enable the vehicle tracking manually. In general, the SO is notified of a missing vehicle by the vehicle owner. The SO then sends a vehicle tracking command message to the TCU to enable vehicle tracking. Periodic messages are sent by the TCU to the SO with location information used in the tracking of the vehicle. A vehicle tracking command message can be sent from the SO to the TCU to disable tracking when it is determined that it is no longer needed. [1]

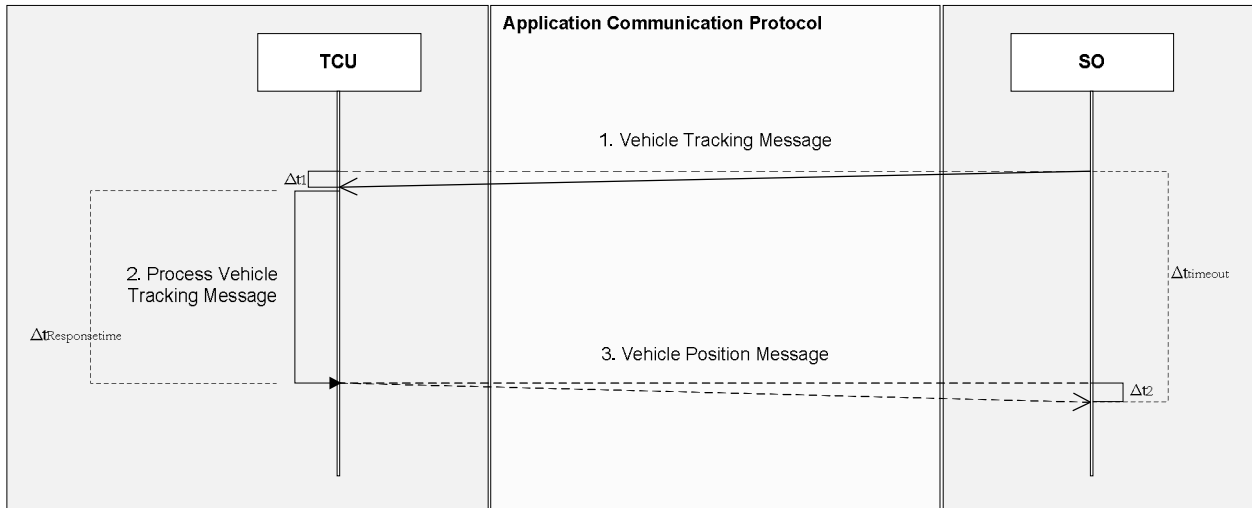


Figure 4 - Vehicle Tracking Message Flow.

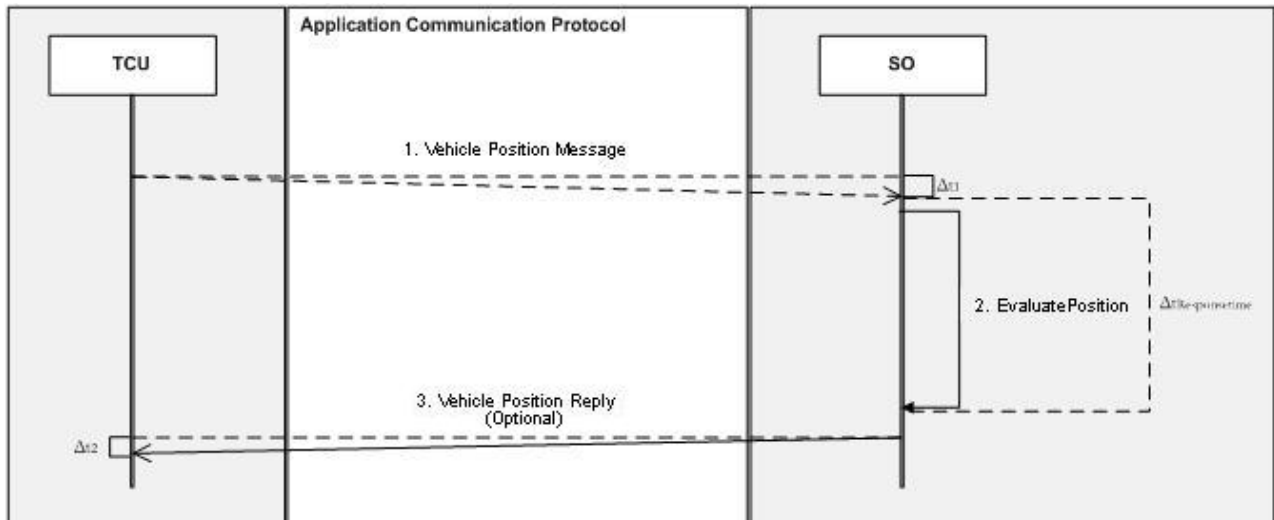


Figure 4.1 – Optional Vehicle Tracking Message Flow.

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

8.1 Vehicle Tracking Message Set

Message Type Id	Message Type
01	Vehicle Tracking Command (*)
02	Vehicle Position Message
03	Vehicle Position Reply
04	Vehicle Tracking Command with Commit (*)
05	Commit to Vehicle Tracking Command (*)

Remarks:

1. (*)Only for ACP 3.1.0.2.
2. Message types 02 and 03 are required for ACP 245.

8.2 Vehicle Tracking Command (From SO to TCU)

Not used in ACP 245. In ACP 245 use “Remote Vehicle Function Command” (Application=6, Message Type=2) for Start/Stop Tracking.

8.3 Vehicle Position Message (From TCU to SO)

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION <i>Message Fields</i>	Header							
	Version Element							
	Timestamp Element							
	Location Element							
	Vehicle Descriptor Element							
	Breakdown Status Element							
	Information Type							

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard Flag	Application ID (default=0x0A)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type(default=0x02)				
	Version Flag	Application Version			Message Control Flag (default=2)			
	Message Length							

Version Element	IE Identifier=0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Timestamp	Year (00=1990, 01=1991 ranges up to 62=2052)		Month of year range (1...12)	
	Month of year	Day of month, range (1...31)		Hour of day
	Hour of day, range (0...23)		Minutes, range (0...59)	
	Minutes	Seconds, range (0...59)		

Location Element	IE Identifier=0	More Flag	Length
	Current GPSRawData (3.8.1)		
	Prior GPSRawData (3.8.1)		
	Current Dead Reckoning Data (3.8.16)		
	Array of Area Location Delta Coding (3.8.17)		

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

Breakdown Status Element	IE Identifier		More Flag	Length				
	More Flag =1	Breakdown Source (<i>First flag</i>)						
	More Flag =1	Breakdown Source (<i>Second flag</i>)						
	More Flag	Breakdown Source (<i>Third flag</i>)						
	More Flag	Breakdown Sensor						
	IE Identifier		More Flag	Length				
	Breakdown Data							

Information Type Element	IE Identifier=0		More Flag	Length	
	Add Flag	Information Type			
	IE Identifier		More Flag	Length	
	Raw Data				

Example: See appendix II at section **Erro! Fonte de referência não encontrada..**

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

8.3.1 Message Elements Definition

8.3.1.1 Header Element

See definitions at section 4.1.

8.3.1.2 Version Element

See definitions at section 3.1.

8.3.1.3 Location Element

See definitions at section 3.8.

8.3.1.4 Vehicle Descriptor Element

See definitions at section 3.4.

8.3.1.5 Breakdown Status Element

See definitions at section 3.9.

8.3.1.6 Information Type Element

See definitions at section 3.10.

8.4 Vehicle Position Reply Message (From SO to TCU)

This message is sent by SO when a request is done by TCU (see item 4.1.7)

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION Message Fields	Header							
	Version Element							
	Confirmation				Transmit Units			
	Ecall ControlFlag2							
	Error Element							

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard Flag	Application ID (default=0x0A)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type (default=0x03)				
	Version Flag	Application Version			Message Control Flag (default=2)			
	Message Length							

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Version Element	IE Identifier=0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Message Fields	Confirmation	Transmit Units
	Ecall ControlFlag2	

Error Element	IE Identifier =0	More Flag	Length
	Error Code		

8.4.1 Message Elements Definition

8.4.1.1 Header Element

See definitions at section 4.1.

8.4.1.2 Version Element

See definitions at section 3.1.

8.4.1.3 Message Fields Element

8.4.1.3.1 Confirmation

This element represents the confirmation status of an assistance reply message.
Default values for ACP 245 – eCall Application not implemented = 0 1 0 1

Bit	Confirmation Definitions
0	Additional Flag
1	1=Assistance notification accepted and processing continues (default) 0=Assistance notification rejected.
2	1=Turn Speaker On 0=Turn Speaker Off (default)
3	1=Processing [tracking] should start or continue (default) 0=Processing [tracking] should stop

Confirmation			
BIT 0	BIT 1	BIT 2	BIT 3

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

8.4.1.3.2 Transmit Units

The value in this element is only valid if the *Confirmation* element indicates tracking should start or continue.

Value	Transmit Units
0	Second (default)
1	Minute
2	Hour
3	Send emergency call message one more time
4	Send only one message

8.4.1.3.3 EcallControlFlag2

This element defines additional control functions in the ecall message.

Bit	Confirmation Definitions
0	Additional Flag
1	1=Cancel the alarm 0=No alarm related action (default)
2	1=Re-send the Ecall request 0=No re-send requested (default)
3	1=Do not allow voice call or drop current voice call (default) 0=Allow voice call (default if EcallControl Flag 2 not present)
4-7	Reserved

eCall Control Flag 2							
BIT 0	BIT 1	BIT 2	BIT 3	Reserved	Reserved	Reserved	Reserved

8.4.1.4 Error Element

See definitions at section 3.5

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9 THEFT ALARM (APPLICATION ID = 11)

The theft alarm application informs the SO that a vehicle event has been triggered. The vehicle manufacturer defines the set of conditions that define a condition event. [1]

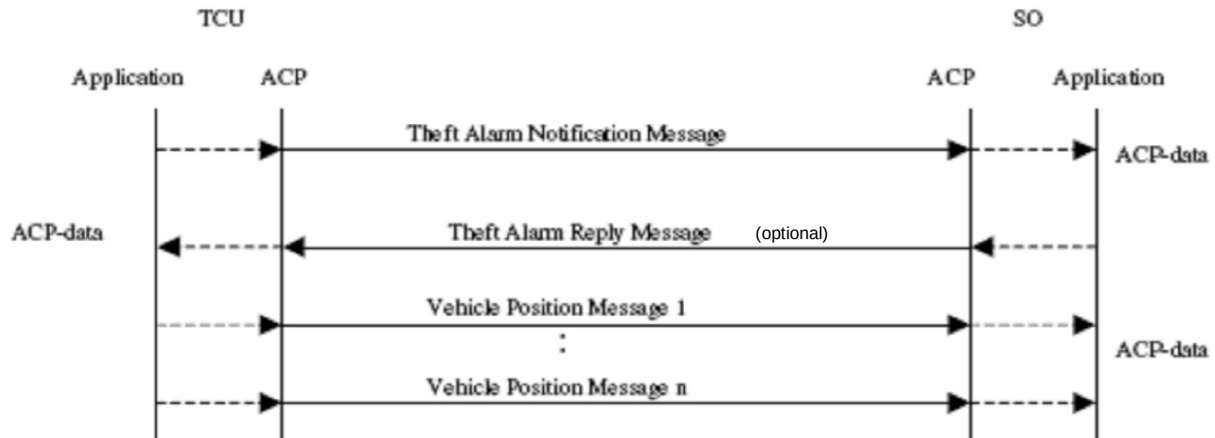


Figure 5 - Theft Alarm Message Flow.

9.1 Theft Alarm Message Set

Message Type Id	Message Type
01	Theft Alarm Notification
02	Theft Alarm Reply
03	Vehicle Position Message
04	Keep Alive
05	Keep Alive Reply

9.2 Theft Alarm Notification (From TCU to SO)

This message is used to inform that an event has occurred and the proper Breakdown Source Flag must be set.

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION Message Fields	Header Element							
	Version Element							
	Timestamp Element							
	Location Element							
	Vehicle Descriptor Element							
	Breakdown Status Element							
	Information Type Element							

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard Flag	Application ID (default=0x0B)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type (default=0x01)				
	Version Flag	Application Version			Message Control Flag (default=2)			
	Message Length							

Version Element	IE Identifier=0	More Flag	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Timestamp	Year (00=1990, 01=1991 ranges up to 62=2052)			Month of year range (1...12)	
	Month of year	Day of month, range (1...31)			Hour of day
	Hour of day, range (0...23)		Minutes, range (0...59)		
	Minutes	Seconds, range (0...59)			

Location Element	IE Identifier=0	More Flag	Length
	Current GPSRawData (3.8.1)		
	Prior GPSRawData (3.8.1)		
	Current Dead Reckoning Data (3.8.16)		
	Array of Area Location Delta Coding (3.8.17)		

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Breakdown Status Element	IE Identifier=0		More Flag	Length
	More Flag =1	Breakdown Source (<i>First flag</i>)		
	More Flag =1	Breakdown Source (<i>Second flag</i>)		
	More Flag	Breakdown Source (<i>Third flag</i>)		
	More Flag	Breakdown Sensor		
	IE Identifier		More Flag	Length
	Breakdown Data			

Information Type Element	IE Identifier=0		More Flag	Length
	Add Flag	Information Type		
	IE Identifier		More Flag	Length
	Raw Data			

9.2.1 Message Elements Definition

9.2.1.1 Header Element

See definitions at section 4.1.

9.2.1.2 Version Element

See definitions at section 3.1.

9.2.1.3 Timestamp Element

See definitions at section 3.2.

9.2.1.4 Location Element

See definitions at section 3.8.

9.2.1.5 Vehicle Descriptor Element

See definitions at section 3.4.

9.2.1.6 Breakdown Status Element

See definitions at section 3.9.

9.2.1.7 Information Type Element

See definitions at section 3.10.

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

9.3 Theft Alarm Reply (From SO to TCU)

This message is sent by SO when a request is done by TCU (see item 4.1.7)

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION	Header Element							
	Version Element							
	Confirmation				Transmit Units			
	Ecall ControlFlag2							
	Error Element							

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header	Reserved Set to 0	Non Standard Flag	Application ID (default=0x0B)					
	Reserved Set to 0	Non Standard Flag	Test Flag	Message Type				
	Version Flag	Application Version			Message Control Flag			
	Message Length							

Version Element	IE Identifier =0	More Flag (default=0)	Length
	Car Manufacturer ID		
	TCU Manufacturer ID		
	Major hardware release		
	Major software release		

Message Fields	Confirmation	Transmit Units
	EcallControlFlag2	

Error Element	IE Identifier =0	More Flag (default=0)	Length
	Error Code		

9.3.1 Message Elements Definition

9.3.1.1 Header Element

See definitions at section 4.1.

9.3.1.2 Version Element

See definitions at section 3.1.

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9.3.1.3 Message Fields Element

See definitions at section 8.4.1.

9.3.1.4 Error Element

See definitions at section 3.5.

9.4 Vehicle Position Message (TCU to SO)

See definitions at section 8.3

9.5 Message Keep Alive (TCU to SO)

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION <i>Message Fields</i>	Header Element							
	Vehicle Descriptor Element							

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header Element	Reserved Set to 0	Private Flag	Application ID (default = 0x0B)					
	Reserved Set to 0	Private Flag	Test Flag	Message Type (default = 0x04)				
	Version Flag	Version			Message Control Flag			
	Message Length							

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

9.5.1 Header Element

See definitions at section 4.1.

9.5.2 Vehicle Descriptor Element

See definitions at section 3.4.

9.6 Message Keep Alive Reply (SO to TCU)

This message is sent by SO when a request is done by TCU (see item 4.1.7)

Message Definition:

Octet / Bit	0	1	2	3	4	5	6	7
MESSAGE COMPOSITION Message Fields	Header Element							
	Vehicle Descriptor Element							

Message Structure:

Octet / Bit	0	1	2	3	4	5	6	7
Header Element	Reserved Set to 0	Private Flag	Application ID (default = 0x0B)					
	Reserved Set to 0	Private Flag	Test Flag	Message Type (default = 0x05)				
	Version Flag	Version			Message Control Flag			
	Message Length							

Vehicle Descriptor Element	IE Identifier=0		More Flag	Length				
	Addl Flag=1	Language	VIN	TCU Serial	Vehicle Color	Vehicle Model	License Plate	IMEI
	Addl Flag=0	Model Year	SIMCard ID	Auth. Key	0	0	0	0
	IE Identifier=1		More Flag	Length				
	Text Format (VIN Number)							
	IE Identifier		More Flag	Length				
	(TCU Serial Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (IMEI Number)							
	IE Identifier=2		More Flag	Length				
	BCD Format (SIM Card ID)							
	IE Identifier=0		More Flag	Length				
	Binary Format (Auth. Key)							

9.6.1 Header Element

See definitions at section 4.1.

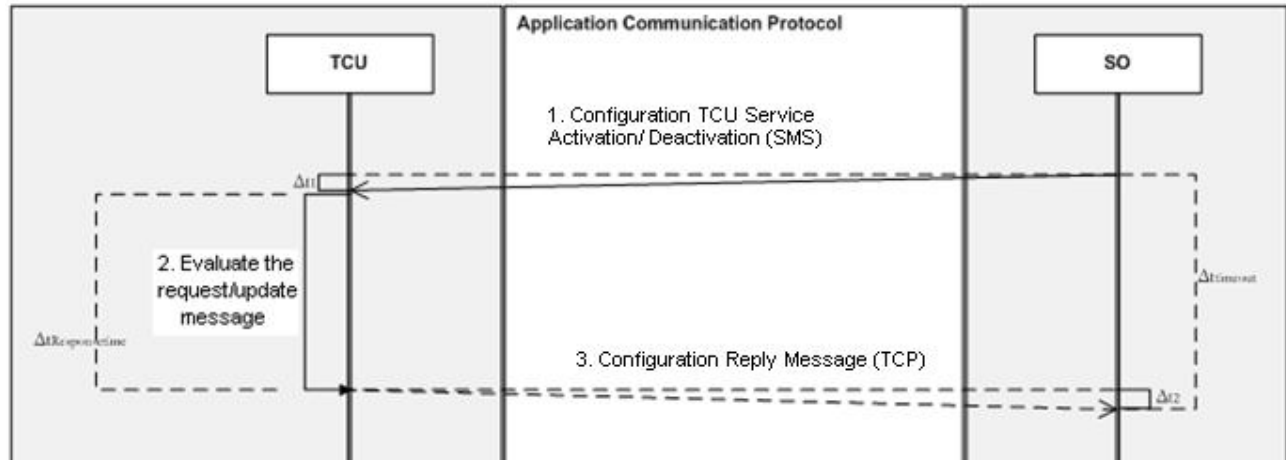
9.6.2 Vehicle Descriptor Element

See definitions at section 3.4.

10 TCU Service Activation/ Deactivation

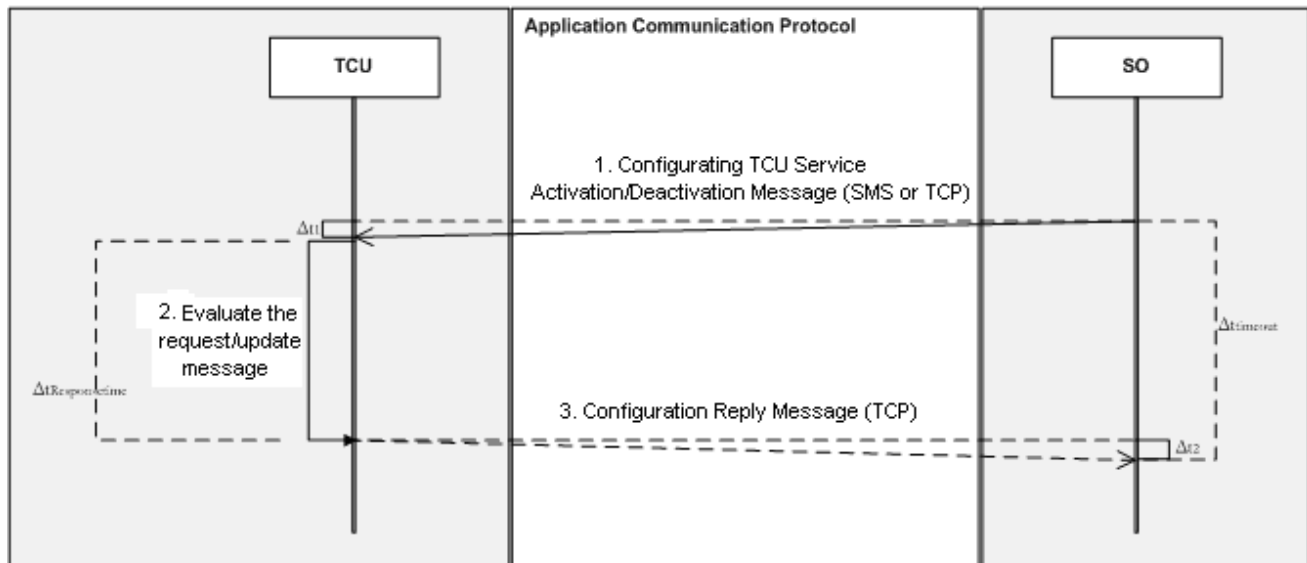
10.1 Activation process

This picture below shows the message flow between SO and TCU in TCU Service Activation process.



10.2 Deactivation process

This picture below shows the message flow between SO and TCU in SMS deactivation process.



11 REFERENCES

- [1] *Application Communication Protocol. Application Layer Message Set Definition*. Version 3.1.0.2. Motorola, Inc. November 2000.
- [2] *DENATRAN Resolution n°245/2007*. July 27, 2007.

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

12 Appendix I

12.1 Available Configuration Parameter Indexes

The table below

12.1.1 Tracking Service

Available indexes to change configuration parameters of tracking service.

Index below marked with * must be configurable in TCU (mandatory).

Data Type Index	Parameter	Description	Type/Size	Default Unit	Default Value
* 0x0011	Tracking Timer	Transmission interval used to send periodically the Vehicle's position. (value 0 – disables tracking)	unsigned integer/ 2 Bytes	Seconds	420
0x0012	Tracking Timer at Sleep Mode	Transmission interval used to send periodically the Vehicle's position, now in Sleep Mode.	unsigned integer/ 2 Bytes	Minutes	1440
0x0013	Type of GPS Quality	Consists of the type of GPS Signal Indication: BIT 0 = Digital indication (0=bad, 1=good) BIT 1 = DOP (Dilution of Precision) in upper nibble and # of satellites in lower nibble of state table.	unsigned char/1 Byte	Enumeration Type	1
0x0014	Direction Change Threshold	Limit used by GPS to detect a direction change	unsigned char/ 1 Byte	degrees	45
0x0015	Direction Change Speed Threshold	Limit used by GPS to alarm detect a direction change	unsigned char/ 1 Byte	Km/h	250
0x0016	Maximum Speed	Maximum Speed to monitoring	unsigned char/ 1 Byte	Km/h	250
0x0017	Odometer Limit	Limit used to count a maximum odometer's values	unsigned long/ 4 Bytes	Km	0 (Unlimited)
0x0018	Odometer Save Threshold	Value used to consider to the odometer counting	unsigned integer/ 2 Bytes	Km	10
0x0019	Tracking Timer at Event Mode	Transmission interval used to send periodically the Vehicle's position, now in Event Mode	unsigned integer/ 2 Bytes	Seconds	300

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12.1.2 Immobilizer Service

Available indexes to change configuration parameters of immobilizer service.

Data Type Index	Parameter	Description	Type	Default Unit	Default Value
0x0021	Immobilizer Timer	Necessary timer used to activate the immobilizer device after turn off the car's ignition.	unsigned integer/ 2 Bytes	Seconds	300

12.1.3 Anti-Theft Service

Available indexes to change configuration parameters of anti-theft service.

Data Type Index	Parameter	Description	Type	Default Unit	Default Value
0x0031	Anti-Theft Timer	Necessary timer used to activate the anti-theft system after ignition is turn off.	unsigned integer/ 2 bytes	Seconds	30
0x0032	Enable/Disable Anti-Theft	Enable/disable the Anti-Theft system.	Boolean/ 1 bit	-	True

12.1.4 System Service

Available indexes to change configuration parameters of system service.

Data Type Index	Parameter	Description	Type	Default Unit	Default Value
0x0041	Sleep Timer	Necessary timer used to put the system in Sleep Mode after the ignition key is turn off.	unsigned integer/ 2 bytes	Seconds	60
0x0042	Enable Panic Button Configuration	Enable a system's input to work like panic button or anti-theft.	Boolean/1 bit	-	False (Anti-Theft is enabled)
0x0043	Enable Transportation Flag.	If this flag is enabled the system will be stay in Transportation Mode.	Boolean/1 bit	-	True

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0x0044	Enable Contract Flag.	Flag used to enable/disable the contract with the SO. Initially and currently only using a SMS message it's possible to enable this flag. After enabled is possible disable it through a GSM/GPRS connection.	Boolean/1 bit	-	False
0x0045	Auth. Key	Authentication key used for SMS validation.	unsigned integer/8 bytes		
*0x0046	TCU Service Activation/ Deactivation	Enable/ Disable the SMS to TCU Service Activation/ Deactivation	Boolean/1 bit	-	True
*0x0047	Vehicle Location Data	Enable/ Disable TCU to send Vehicle Location Data (Location Element with Length "0")	Boolean/1 bit	-	False

12.1.5 Network Service

Available indexes to change configuration parameters of network service.

Data Type Index	Parameter	Description	Type	Default Unit	Default Value
0x0051	Keep Alive Timer	Necessary timer to send a small message to verify the connection status between the TCU and the SO. (value 0 – disables)	unsigned int / 2 byte	Seconds	180
0x0052	Default Transport Protocol	Default transport protocol used to establish connection between the TCU and the SO.	unsigned char/ 1 byte	-	0 (TCP)

12.1.6 Connectivity Service

Available indexes to change configuration parameters of connectivity service.

Data Type Index	Parameter	Description	Type	Default Unit	Default Value
* 0x0061	APN (Access Point Name)	Access point used to start or to establish a connection.	Max. 91 Bytes (Sum)	-	-
* 0x0062	Login	Login or user name used to access the services provided by SO (Service Operator)		-	-
* 0x0063	Password	Password used to access the services provided by SO (Service Operator)		-	-
* 0x0064	Server's IP # 1	First IP used to establish a connection in the SO network.	4 Bytes	-	0.0.0.0
* 0x0065	Server's Port # 1	First port number used to establish data transmission between client/server in the SO network.	unsigned integer /2 Bytes	-	0

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0x0066	Server's Transport Protocol Type # 1	Type of the transport protocol used (currently: TCP = 0, or UDP =1)	unsigned char/ 1 Byte	-	0 (TCP)
0x0067	Server's IP # 2	Second IP used to establish a connection in the SO network.	4 Bytes	-	0.0.0.0
0x0068	Server's Port # 2	Second port number used to establish data transmission between client/server in the SO network.	unsigned integer /2 Bytes	-	0
0x0069	Server's Transport Protocol Type # 2	Type of the transport protocol used (currently: TCP = 0, or UDP =1)	unsigned char/ 1 Byte	-	0 (TCP)
0x006A	Server's IP # 3	Third IP used to establish a connection in the SO network.	4 Bytes	-	0.0.0.0
0x006B	Server's Port # 3	Third port number used to establish data transmission between client/server in the SO network.	unsigned integer /2 Bytes	-	0
0x006C	Server's Transport Protocol Type # 3	Type of the transport protocol used (currently: TCP = 0, or UDP =1)	unsigned char/ 1 Byte	-	0 (TCP)
0x006D	Server's IP # 4	Fourth IP used to establish a connection in the SO network.	4 Bytes	-	0.0.0.0
0x006E	Server's Port # 4	Fourth port number used to establish data transmission between client/server in the SO network.	unsigned integer /2 Bytes	-	0
0x006F	Server's Transport Protocol Type # 4	Type of the transport protocol used (currently: TCP = 0, or UDP =1)	unsigned char/ 1 Byte	-	0 (TCP)
0x0070	Server's IP # 5	Fifth IP used to establish a connection in the SO network.	4 bytes	-	0.0.0.0
0x0071	Server's Port # 5	Fifth port number used to establish data transmission between client/server in the SO network.	unsigned integer /2 Bytes	-	0
0x0072	Server's Transport Protocol Type # 5	Type of the transport protocol used (currently: TCP = 0, or UDP =1)	unsigned char/ 1 Byte	-	0 (TCP)
0x0073	Server's IP # 6	Sixth IP used to establish a connection in the SO network.	4 bytes	-	0.0.0.0
0x0074	Server's Port # 6	Sixth port number used to establish data transmission between client/server in the SO network.	unsigned integer /2 Bytes	-	0
0x0075	Server's Transport Protocol Type # 6	Type of the transport protocol used (currently: TCP = 0, or UDP =1)	unsigned char/ 1 Byte	-	0 (TCP)

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12.1.7 Power Service

Available indexes to change configuration parameters of power service.

Data Type Index	Parameter	Description	Type	Default Unit	Default Value
0x0081	Minimal Voltage for the Main Battery.	Minimal voltage value allowed to operate the system with Main Battery	unsigned integer/ 2 Bytes	-	109
0x0082	Maximum Voltage for the Main Battery	Maximum voltage value allowed to operate the system with Main Battery	unsigned integer/ 2 Bytes	-	727
0x0083	Hysteresis value for the Main Battery's voltage	Hysteresis value used to calculate the Main Battery's voltage.	unsigned integer/ 2 Bytes	-	5
0x0084	Minimal Voltage for the Backup Battery.	Minimal voltage value allowed to operate the system with Backup Battery	unsigned integer/ 2 Bytes	-	750
0x0085	Recharge Voltage for the Backup Battery	Voltage value used to start the backup battery's recharge process when in active or inactive mode.	unsigned integer/ 2 Bytes	-	970
0x0086	Charged Voltage for the Backup Battery's	Voltage value used to stop the backup battery's recharge process when in active or inactive mode.	unsigned integer/ 2 Bytes	-	995
0x0087	Hysteresis value for the Backup Battery's voltage	Hysteresis value used to calculate the Backup Battery's voltage.	unsigned integer/ 2 Bytes	-	20
0x0088	Backup Battery's Minimal Temperature	Minimal temperature value used to start/continue the backup battery's recharge process.	unsigned integer/ 2 Bytes	-	697
0x0089	Backup Battery's Maximum Temperature	Maximum temperature value used to start/continue the backup battery's recharge process.	unsigned integer/ 2 Bytes	-	444
0x008A	Hysteresis value for the Backup Battery's temperature	Hysteresis value used to calculate the Backup Battery's temperature.	unsigned integer/ 2 Bytes	-	111
0x008B	Backup Battery's Safety Timer 1	Consist of the first safety timer used to check and validate the recharge process.	unsigned integer/ 2 Bytes	Seconds	90
0x008C	Backup Battery's Safety Timer 2	Consist of the second safety timer used to check and validate the recharge process. This value also establish the maximum backup battery's recharge time.	unsigned integer/ 2 Bytes	Seconds	3300
0x008D	Maximum	Maximum number to allow a fail	unsigned char/	-	3

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	Backup Battery's Fail Counter	during the recharge process. That fail occur if even reached the Safety Timer 2, the backup battery's voltage still below of its charged value.	1 Byte		
0x008E	Backup Battery's Validity Year	Value used to establish the validity year of the backup battery start year = 2000 = 00	unsigned char/ 1 Byte	-	12 (2012)
0x008F	Backup Battery's Validity Month	Value used to establish the validity month of the backup battery range: 01 – 31	unsigned char/ 1 Byte	-	01 (January)

12.1.8 Alarm Service

Available indexes to change configuration parameters of alarm service.

Data Type Index	Parameter	Description	Type	Default Unit	Default Value
0x0092	Ignition Event – Activation time	Time necessary to activate any output after the ignition event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x0095	Panic Button Event – Activation time	Time necessary to activate any output after the panic button event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x0098	GPS Malfunctioning Event – Activation time	Time necessary to activate any output after the GPS Malfunctioning event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x009B	Main Battery Low Event – Activation time	Time necessary to activate any output after the Main Battery Low event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x009E	Backup Battery Temperature Event – Activation time	Time necessary to activate any output after the Backup Battery's Temperature event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00A2	Anti-Theft Violation Event – Activation time	Time necessary to activate any output after the Anti-Theft Violation event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x01
0x00A5	Sleep Event –	Time necessary to activate any output after the Sleep event to be	unsigned	0,1 Seconds	0x0000 – Event not

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	Activation time	considered stable and ready to transmit.	char/1 Byte		allowed to transmit (Event Disable).
0x00A8	Position Violation Event – Activation time	Time necessary to activate any output after the Position Violation event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00AB	Weak GSM Signal Event – Activation time	Time necessary to activate any output after the Weak GSM Signal event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00AE	Backup Battery Low Event – Activation time	Time necessary to activate any output after the Backup Battery Low event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00B2	Backup Battery Fail Event – Activation time	Time necessary to activate any output after the Backup Battery Fail event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00B5	Backup Battery End of Life Event – Activation time	Time necessary to activate any output after the Backup Battery End of Life event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00B8	Output Short Circuit Event – Activation time	Time necessary to activate any output after the Output Short Circuit event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00BB	Maximum Speed Event – Activation time	Time necessary to activate any output after the Maximum Speed event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00BE	Course Change Event – Activation time	Time necessary to activate any output after the Course Change event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00C2	Number GPS of satellite drop Event – Activation time	Time necessary to activate any output after the Number GPS of satellite drop event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).

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0x00C5	GSM Antenna Cutoff Event – Activation time	Time necessary to activate any output after the GSM Antenna Cutoff event to be considered stable and ready to transmit.	unsigned char/1 Byte	0,1 Seconds	0x0000 – Event not allowed to transmit (Event Disable).
0x00C6	Enable/Disable Alarm Services	Enable or disable the notification of alarms and/or events.	Boolean/1 bit	-	True

12.1.9 FOTA Service

Data Type Index	Parameter	Description	Type	Default Unit	Default Value
0x00D1	Enable/Disable FOTA Service	Enable or disable the service used to Firmware Update Over The Air.	Boolean/1 bit	-	True

12.1.10 Custom parameters

Data Type Index	Parameter	Description	Type	Default Unit	Default Value
0x0100 to 0x01FF	Reserved	Reserved for custom application	NA	NA	NA

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13 Appendix II – Telematics Operation – Minimum Message Set Guideline

The Appendix II was created to establish a minimum common set of messages to be used in Telematics applications using ACP v1.2.2.

13.1 General Information

- 1) In the examples of messages sent from TCU to SO, the Version Element is included because it's required in the process of connections and reconections (see Appendix III – Item 14.3).
- 2) In the examples of messages sent from SO to TCU, the Version Element is not included because it's not required.

13.2 Example: Converting Position Coordinates

Longitude

Example: F5E61D8D = - 169468531 milliarcseconds = - 47.074592 degrees

(F5E61D8D) 2's comp hexadecimal milliarcsec

(1111 0101 1110 0110 0001 1101 1000 1101) 2's comp binary milliarcsec

- (0000 1010 0001 1001 1110 0010 0111 0011) binary milliarcsec

- (169468531) milliarcsec

- (169468.531) seconds

- (2824.475516666666667) minutes

- (47.074591944444444444) degrees

-47.074592

Latitude

Example: FB19EDCB = - 82186805 milliarcseconds = - 22.829668 degrees

(FB19EDCB) 2's comp hexadecimal milliarcsec

(1111 1011 0001 1001 1110 1101 1100 1011) 2's comp binary milliarcsec

- (0000 0100 1110 0110 0001 0010 0011 0101) binary milliarcsec

- (82186805) milliarcsec

- (82186.805) seconds

- (1369.780083333333333) minutes

- (22.82966805555555556) degrees

-22.829668

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13.3 PROVISIONING

13.3.1 Activation attempt without version element

13.3.1.1 Provision Update Message #1 (From SO to TCU)

Provision Update Message #1 (Activation attempt)
Raw Message (Hexadecimal Format)
01011116000B42000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x01	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x01 (Provisioning)
2		0x01	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x01 (Provisioning Update Msg. #1)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x16	Message Length = 0x16 (22)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x0B	More Flag = 0x00; Target Application Id = 0x0B (Theft Alarm);
7		0x42	AppFlag1 = 0x01 (Activate); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8	TCU descriptor Element	0x00	IE Element = 0x00; More Flag = 0x00; Length = 0x00
9	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
10		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
11		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
12		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
13		0x12	0x12
14		0x34	0x34
15		0x56	0x56
16		0x78	0x78
17		0x90	0x90
18		0x12	0x12
19		0x34	0x34
20		0x56	0x56
21		0x78	0x78
22		0x90	0x90

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13.3.1.2 Provision Reply Message #1 (From TCU to SO)

Provision Reply Message #1
Raw Message (Hexadecimal Format)
0103101C04YYZZ01020B422001000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x01	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x01 (Provisioning)
2		0x03	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x03 (Provisioning reply Msg. #1)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x1C	Message Length = 0x1C (28)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10	Message Fields	0x0B	More Flag = 0x00; Target Application Id = 0x0B (Theft Alarm);
11		0x42	AppFlag1 = 0x01 (Activate); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
12		0x20	StatusFlag1 = 0x00 (Application already provisioned); TCU Response Flag = 0x02; Reserved = 0x00
13	Error Element	0x01	IE Element = 0x00; More Flag = 0x00; Length = 0x01
14		0x00	Error Code = 0x00 (OK)
15	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
16		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
17		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
18		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
19		0x12	0x12
20		0x34	0x34
21		0x56	0x56
22		0x78	0x78
23		0x90	0x90
24		0x12	0x12
25		0x34	0x34
26		0x56	0x56
27		0x78	0x78
28		0x90	0x90

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13.3.2 Consult request: no Version Element

13.3.2.1 Provision Update Message #1 (From SO to TCU)

Provision Update Message #1
Raw Message (Hexadecimal Format)
01011116000B02000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x01	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x01 (Provisioning)
2		0x01	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x01 (Provisioning Update Msg. #1)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x16	Message Length = 0x16 (22)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x0B	More Flag = 0x00; Target Application Id = 0x0B (Theft Alarm);
7		0x02	AppFlag1 = 0x00 (No change); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8	TCU descriptor Element	0x00	IE Element = 0x00; More Flag = 0x00; Length = 0x00
9	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
10		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
11		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
12		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
13		0x12	0x12
14		0x34	0x34
15		0x56	0x56
16		0x78	0x78
17		0x90	0x90
18		0x12	0x12
19		0x34	0x34
20		0x56	0x56
21		0x78	0x78
22		0x90	0x90

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13.3.2.2 Provision Reply Message #1 (From TCU to SO)

Provision Reply Message #1
Raw Message (Hexadecimal Format)
01031018000B022001000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x01	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x01 (Provisioning)
2		0x03	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x03 (Provisioning reply Msg. #1)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x18	Message Length = 0x18 (24)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x0B	More Flag = 0x00; Target Application Id = 0x0B (Theft Alarm);
7		0x02	AppFlag1 = 0x00 (No change); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8		0x20	StatusFlag1 = 0x00 (Application already provisioned); TCU Response Flag = 0x02; Reserved = 0x00
9	Error Element	0x01	IE Element = 0x00; More Flag = 0x00; Length = 0x01
10		0x00	Error Code = 0x00 (OK)
11	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
12		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
13		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
14		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
15		0x12	0x12
16		0x34	0x34
17		0x56	0x56
18		0x78	0x78
19		0x90	0x90
20		0x12	0x12
21		0x34	0x34
22		0x56	0x56
23		0x78	0x78
24		0x90	0x90

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13.3.3 Deactivation attempt (invalid): no Version Element, “Tracking” deactivation

13.3.3.1 Provision Update Message #1 (From SO to TCU)

Provision Update Message #1
Raw Message (Hexadecimal Format)
01011116000A82000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x01	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x01 (Provisioning)
2		0x01	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x01 (Provisioning Update Msg. #1)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x16	Message Length = 0x16 (22)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x0A	More Flag = 0x00; Target Application Id = 0x0A (Tracking);
7		0x82	AppFlag1 = 0x02 (Deactivate); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8	TCU descriptor Element	0x00	IE Element = 0x00; More Flag = 0x00; Length = 0x00
9	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
10		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
11		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
12		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
13		0x12	0x12
14		0x34	0x34
15		0x56	0x56
16		0x78	0x78
17		0x90	0x90
18		0x12	0x12
19		0x34	0x34
20		0x56	0x56
21		0x78	0x78
22		0x90	0x90

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13.3.3.2 Provision Reply Message #1 (From TCU to SO)

Provision Reply Message #1
Raw Message (Hexadecimal Format)
01031018000A02A001130D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x01	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x01 (Provisioning)
2		0x03	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x03 (Provisioning reply Msg. #1)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x18	Message Length = 0x18 (24)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x0A	More Flag = 0x00; Target Application Id = 0x0A (Tracking);
7		0x02	AppFlag1 = 0x00 (No change); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8		0xA0	StatusFlag1 = 0x02 (See Error Element); TCU Response Flag = 0x02; Reserved = 0x00
9	Error Element	0x01	IE Element = 0x00; More Flag = 0x00; Length = 0x01
10		0x13	Error Code = 0x13 (19-General non-specific error)
11	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
12		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
13		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
14		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
15		0x12	0x12
16		0x34	0x34
17		0x56	0x56
18		0x78	0x78
19		0x90	0x90
20		0x12	0x12
21		0x34	0x34
22		0x56	0x56
23		0x78	0x78
24		0x90	0x90

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13.4 CONFIGURATION FOR TELEMATICS SERVICES

13.4.1 Updating Tracking Timer without version element

13.4.1.1 Configuration Update Message #2 ACP 245 (From SO to TCU)

Configuration Update Message #2 (Updating Tracking Timer)
Raw Message (Hexadecimal Format)
0208111D0000C2000D80208A123456789012345678900005001102003C

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x08	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x08 (Configuration Update Msg. #2)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x1D	Message Length = 0x1D (29)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x00	More Flag = 0x00; Target Application Id = 0x00;
7		0xC2	AppFlag1 = 0x03 (Change for this Application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8		0x00	Reserved = 0x00
9	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
10		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
11		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
12		0x8A	IE Identifier = 0x02(BCD Coding); More Flag = 0x00; Length = 0x0A
13		0x12	0x12
14		0x34	0x34
15		0x56	0x56
16		0x78	0x78
17		0x90	0x90
18		0x12	0x12
19		0x34	0x34
20		0x56	0x56
21		0x78	0x78
22		0x90	0x90
23	TCU Descriptor Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00

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24	TCU Data Element	0x05	IE Identifier = 0x00; More Flag = 0x00; Length = 0x05
25		0x00	Data Type (MSB) = 0x00
26		0x11	Data Type (LSB) = 0x11 (Tracking Timer)
27		0x02	Length Data Type = 0x02
28		0x00	Configuration Data [0] = 0x00
29		0x3C	Configuration Data [1] = 0x3C (60 Seconds)

13.4.1.2 Configuration Reply Message #2 ACP 245 (From TCU to SO)

Configuration Reply Message #2
Raw Message (Hexadecimal Format)
0209102204YYZZ010200C20X07001102003C01000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail
1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x09	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x09 (Configuration Reply Msg. #2)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x22	Message Length = 0x22 (34)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10	Message Fields	0x00	More Flag = 0x00; Target Application Id = 0x00;
11		0xC2	AppFlag1 = 0x03 (Change for this Application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
12		0x0X	Status Flag1 = 0x0X(Accepted values = 0,2,3);TCU Response Flag = 0x00; Reserved = 0x00
13	TCU Data Error Element	0x07	IE Identifier = 0x00; More Flag = 0x00; Length = 0x07
14		0x00	Data Type (MSB) = 0x00
15		0x11	Data Type (LSB) = 0x11 (Tracking Timer)
16		0x02	Length Data Type = 0x02
17		0x00	Configuration Data [0] = 0x00
18		0x3c	Configuration Data [1] = 0x3C (60 Seconds)
19		0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
20		0x00	Error Code = 0x00 (Ok)
21	Vehicle Descriptor	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
22		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)

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23	Element	0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
24		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
25		0x12	0x12
26		0x34	0x34
27		0x56	0x56
28		0x78	0x78
29		0x90	0x90
30		0x12	0x12
31		0x34	0x34
32		0x56	0x56
33		0x78	0x78
34		0x90	0x90

13.4.2 Updating Multiple Parameters without version element

Parameters: Tracking timer, Tracking timer at event mode, Keep alive timer.

13.4.2.1 Configuration Update Message #2 ACP 245 (From SO to TCU)

Configuration Update Message #2
Raw Message (Hexadecimal Format)
020811270000C2000D80208A12345678901234567890000F001102003C001902001E0051020096

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x08	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x08 (Configuration Update Msg. #2)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x27	Message Length = 0x27 (39)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x00	More Flag = 0x00; Target Application Id = 0x00;
7		0xC2	AppFlag1 = 0x03(Change for this Application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8		0x00	Reserved = 0x00
9	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
10		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
11		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
12		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00;

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			Length = 0x0A
13		0x12	0x12
14		0x34	0x34
15		0x56	0x56
16		0x78	0x78
17		0x90	0x90
18		0x12	0x12
19		0x34	0x34
20		0x56	0x56
21		0x78	0x78
22		0x90	0x90
23	TCU Descriptor Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
24	TCU Data Element	0x0F	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0F
25		0x00	Data Type (MSB) = 0x00
26		0x11	Data Type (LSB) = 0x11 (Tracking Timer)
27		0x02	Length Data Type = 0x02
28		0x00	Configuration Data [0] = 0x00
29		0x3C	Configuration Data [1] = 0x3C (60 Seconds)
30		0x00	Data Type (MSB) = 0x00
31		0x19	Data Type (LSB) = 0x19 (Tracking Timer at Event Mode)
32		0x02	Length Data Type = 0x02
33		0x00	Configuration Data [0] = 0x00
34		0x1E	Configuration Data [1] = 0x1E (30 Seconds)
35		0x00	Data Type (MSB) = 0x00
36		0x51	Data Type (LSB) = 0x51 (Keep Alive Timer)
37		0x02	Length Data Type = 0x02
38		0x00	Configuration Data [0] = 0x00
39		0x96	Configuration Data [1] = 0x96 (150 Seconds)

13.4.2.2 Configuration Reply Message #2 ACP 245 (From TCU to SO)

Configuration Reply Message #2	
Raw Message (Hexadecimal Format)	
0209103004YYZZ010200C20X15001102003C0100001902001E0100005102009601000D80208A12345678901234567890	

Message Detail (Message Elements)			
Octet	Element	Value	Detail
1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)

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2		0x09	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x09 (Configuration Reply Msg. #2)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x30	Message Length = 0x30 (48)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10	Message Fields	0x00	More Flag = 0x00; Target Application Id = 0x00;
11		0xC2	AppFlag1 = 0x03 (Change for this Application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
12		0x0X	Status Flag1 = 0x0X (Accepted values = 0,2,3); TCU Response Flag = 0x00; Reserved = 0x00
13	TCU Data Error Element	0x15	IE Identifier = 0x00; More Flag = 0x00; Length = 0x15
14		0x00	Data Type (MSB) = 0x00
15		0x11	Data Type (LSB) = 0x11 (Tracking Timer)
16		0x02	Length Data Type = 0x02
17		0x00	Configuration Data [0] = 0x00
18		0x3C	Configuration Data [1] = 0x3C (60 Seconds)
19		0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
20		0x00	Error Code = 0x00
21		0x00	Data Type (MSB) = 0x00
22		0x19	Data Type (LSB) = 0x19 (Tracking Timer at Event Mode)
23		0x02	Length Data Type = 0x02
24		0x00	Configuration Data [0] = 0x00
25		0x1E	Configuration Data [1] = 0x1E (30 Seconds)
26		0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
27		0x00	Error Code = 0x00
28		0x00	Data Type (MSB) = 0x00
29		0x51	Data Type (LSB) = 0x51 (Keep Alive Timer)
30		0x02	Length Data Type = 0x02
31		0x00	Configuration Data [0] = 0x00
32		0x96	Configuration Data [1] = 0x96 (150 Seconds)
33		0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
34		0x00	Error Code = 0x00

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

35	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
36		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
37		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
38		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
39		0x12	0x12
40		0x34	0x34
41		0x56	0x56
42		0x78	0x78
43		0x90	0x90
44		0x12	0x12
45		0x34	0x34
46		0x56	0x56
47		0x78	0x78
48		0x90	0x90

13.4.3 Consulting Single Parameter: no version element

Parameter: Tracking timer

13.4.3.1 Configuration Update Message #2 ACP 245 (From SO to TCU)

Configuration Update Message #2
Raw Message (Hexadecimal Format)
0208111B000002000D80208A123456789012345678900003001100

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x08	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x08 (Configuration Update Msg. #2)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x1B	Message Length = 0x1B (27)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x00	More Flag = 0x00; Target Application Id = 0x00;
7		0x02	AppFlag1 = 0x00 (No Change); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8		0x00	Reserved = 0x00
9	Vehicle Descriptor	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

10	Element	0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
11		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
12		0x8A	IE Identifier = 0x02(BCD Coding); More Flag = 0x00; Length = 0x0A
13		0x12	0x12
14		0x34	0x34
15		0x56	0x56
16		0x78	0x78
17		0x90	0x90
18		0x12	0x12
19		0x34	0x34
20		0x56	0x56
21		0x78	0x78
22		0x90	0x90
23	TCU Descriptor Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
24	TCU Data Element	0x03	IE Identifier = 0x00; More Flag = 0x00; Length = 0x03
25		0x00	Data Type (MSB) = 0x00
26		0x11	Data Type (LSB) = 0x11 (Tracking Timer)
27		0x00	Length Data Type = 0x00

13.4.3.2 Configuration Reply Message #2 ACP 245 (From TCU to SO)

Configuration Reply Message #2
Raw Message (Hexadecimal Format)
0209102204YYZZ010200020007001102003C01000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x09	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x09 (Configuration Reply Msg. #2)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x22	Message Length = 0x22 (34)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)

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8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10		0x00	More Flag = 0x00; Target Application Id = 0x00;
11	Message Fields	0x02	AppFlag1 = 0x00 (No change to Application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
12		0x00	Status Flag1 = 0x00; TCU Response Flag = 0x00; Reserved = 0x00
13		0x07	IE Identifier = 0x00; More Flag = 0x00; Length = 0x07
14	TCU Data Error Element	0x00	Data Type (MSB) = 0x00
15		0x11	Data Type (LSB) = 0x11 (Tracking Timer)
16		0x02	Length Data Type = 0x02
17		0x00	Configuration Data [0] = 0x00
18		0x3C	Configuration Data [1] = 0x3C (60 Seconds)
19		0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
20		0x00	Error Code = 0x00
21		0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
22	Vehicle Descriptor Element	0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
23		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
24		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
25		0x12	0x12
26		0x34	0x34
27		0x56	0x56
28		0x78	0x78
29		0x90	0x90
30		0x12	0x12
31		0x34	0x34
32		0x56	0x56
33		0x78	0x78
34		0x90	0x90

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

13.4.4 Consulting Multiple Parameters: no version element

Parameters: Tracking timer, Tracking timer at event mode, Keep alive timer.

13.4.4.1 Configuration Update Message #2 ACP 245 (From SO to TCU)

Configuration Update Message #2
Raw Message (Hexadecimal Format)
02081121000002000D80208A123456789012345678900009001100001900005100

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x08	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x08 (Configuration Update Msg. #2)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x21	Message Length = 0x21 (33)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x00	More Flag = 0x00; Target Application Id = 0x00;
7		0x02	AppFlag1 = 0x00 (No Change) Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8	Vehicle Descriptor Element	0x00	Reserved = 0x00
9		0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
10		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
11		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
12		0x8A	IE Identifier = 0x02(BCD Coding); More Flag = 0x00; Length = 0x0A
13		0x12	0x12
14		0x34	0x34
15		0x56	0x56
16		0x78	0x78
17		0x90	0x90
18		0x12	0x12
19		0x34	0x34
20		0x56	0x56
21		0x78	0x78
22		0x90	0x90
23	TCU Descriptor Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
24	TCU Data Element	0x09	IE Identifier = 0x00; More Flag = 0x00; Length = 0x09
25		0x00	Data Type (MSB) = 0x00

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26		0x11	Data Type (LSB) = 0x11 (Tracking Timer)
27		0x00	Length Data Type = 0x00
28		0x00	Data Type (MSB) = 0x00
29		0x19	Data Type (LSB) = 0x19 (Tracking Timer at Event Mode)
30		0x00	Length Data Type = 0x00
31		0x00	Data Type (MSB) = 0x00
32		0x51	Data Type (LSB) = 0x51 (Keep Alive Timer)
33		0x00	Length Data Type = 0x00

13.4.4.2 Configuration Reply Message #2 ACP 245 (From TCU to SO)

Configuration Reply Message #2	
Raw Message (Hexadecimal Format)	
0209103004YYZZ010200020X15001102003C0100001902001E0100005102009601000D80208A12345678901234567890	

Message Detail (Message Elements)			
Octet	Element	Value	Detail
1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x09	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x09 (Configuration Reply Msg. #2)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x30	Message Length = 0x30 (48)
5		0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10	Message Fields	0x00	More Flag = 0x00; Target Application Id = 0x00;
11		0x02	AppFlag1 = 0x00 (No change to Application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
12		0x0X	Status Flag1 = 0x0X (Accepted values = 0,2,3); TCU Response Flag = 0x00; Reserved = 0x00
13	TCU Data Error Element	0x15	IE Identifier = 0x00; More Flag = 0x00; Length = 0x15
14		0x00	Data Type (MSB) = 0x00
15		0x11	Data Type (LSB) = 0x11 (Tracking Timer)
16		0x02	Length Data Type = 0x02
17		0x00	Configuration Data [0] = 0x00

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18		0x3C	Configuration Data [1] = 0x3C (60 Seconds)
19		0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
20		0x00	Error Code = 0x00
21		0x00	Data Type (MSB) = 0x00
22		0x19	Data Type (LSB) = 0x19 (Tracking Timer at Event Mode)
23		0x02	Length Data Type = 0x02
24		0x00	Configuration Data [0] = 0x00
25		0x1E	Configuration Data [1] = 0x1E (30 Seconds)
26		0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
27		0x00	Error Code = 0x00
28		0x00	Data Type (MSB) = 0x00
29		0x51	Data Type (LSB) = 0x51 (Keep Alive Timer)
30		0x02	Length Data Type = 0x02
31		0x00	Configuration Data [0] = 0x00
32		0x96	Configuration Data [1] = 0x96 (150 Seconds)
33		0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
34		0x00	Error Code = 0x00
35	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
36		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
37		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
38		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
39		0x12	0x12
40		0x34	0x34
41		0x56	0x56
42		0x78	0x78
43		0x90	0x90
44		0x12	0x12
45		0x34	0x34
46		0x56	0x56
47		0x78	0x78
48		0x90	0x90

13.4.4.3 Configuration Update Message #2 (From SO to TCU) (Updating Vehicle Location Data).

Configuration Update Message #2 (Updating Vehicle Location Data)
Raw Message (Hexadecimal Format)
0208111C0000C2000D80208A12345678901234567890000400470101

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Message Detail (Message Elements)			
Octet	Element	Value	Detail
1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x08	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x08 (Configuration Update Msg. #2)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x1C	Message Length = 0x1C(28)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Message Fields	0x00	More Flag = 0x00; Target Application Id = 0x00;
7		0xC2	AppFlag1 = 0x03(Change for this Application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
8		0x00	Reserved = 0x00
9	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
10		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
11		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
12		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
13		0x12	0x12
14		0x34	0x34
15		0x56	0x56
16		0x78	0x78
17		0x90	0x90
18		0x12	0x12
19		0x34	0x34
20		0x56	0x56
21		0x78	0x78
22		0x90	0x90
23	TCU Descriptor Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
24	TCU Data Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
25		0x00	Data Type (MSB) = 0x00
26		0x47	Data Type (LSB) = 0x47 (Vehicle Location Data)
27		0x01	Length Data Type = 0x01
28		0x01	Configuration Data [0] = 0x01 (Enable)

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13.4.4.4 Configuration Reply Message #2 (From TCU to SO) (Reply for Request Vehicle Location Data)

Configuration Reply Message #2 (Reply for Request Vehicle Location Data)
Raw Message (Hexadecimal Format)
0209102104YYZZ010200C20X060047010101000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x09	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x09 (Configuration Reply Msg. #2)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x21	Message Length = 0x21 (33)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10	Message Fields	0x00	More Flag = 0x00; Target Application Id = 0x00;
11		0xC2	AppFlag1 = 0x03 (Change for this Application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
12		0x0X	Status Flag1 = 0x0X(Accepted values = 0,2,3);TCU Response Flag = 0x00; Reserved = 0x00
13	TCU Data Error Element	0x06	IE Identifier = 0x00; More Flag = 0x00; Length = 0x06
14		0x00	Data Type (MSB) = 0x00
15		0x47	Data Type (LSB) = 0x47 (Vehicle Location Data)
16		0x01	Length Data Type = 0x01
17		0x01	Configuration Data [0] = 0x01 (Enabled)
18		0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
19		0x00	Error Code = 0x00
20	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
21		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
22		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
23		0x8A	IE Identifier = 0x02(BCD Coding); More Flag = 0x00; Length = 0x0A
24		0x12	0x12
25		0x34	0x34
26		0x56	0x56
27		0x78	0x78

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28		0x90	0x90
29		0x12	0x12
30		0x34	0x34
31		0x56	0x56
32		0x78	0x78
33		0x90	0x90

13.4.5 Configuration TCU Service Activation/ Deactivation Message ACP 245

13.4.5.1 Configuration TCU Service Activation/ Deactivation Message ACP 245 (From SO to TCU) (Activation)

Configuration TCU Service Activation/ Deactivation Message ACP 245 (Activation)
Raw Message (Hexadecimal Format)
020A113A0F4674696D2E62724374696D4374696D0DBD4F22F015B3BD4F22F115BE00401680308A 1234567890123456789008XXXXXXXXXXXXXXXXXX

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x0A	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x0A (Activation/Deactivation Message)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01 (Response is expected)
4		0x3A	Message Length = 0x3A (58)
5	APN Configuration Element	0x0F	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0F (15)
6		0x46	IE Identifier = 0x01 (Text); More Flag = 0x00; Length = 0x06 (06) - (APN Address Element – Start)
7		0x74	0x74 ('t')
8		0x69	0x69 ('i')
9		0x6D	0x6D ('m')
10		0x2E	0x2E ('.')
11		0x62	0x62 ('b')
12		0x72	0x72 ('r')
13		0x43	IE Identifier = 0x01 (Text); More Flag = 0x00; Length = 0x03 (03) - (APN Login Element – Start)
14		0x74	0x74 ('t')
15		0x69	0x69 ('i')
16		0x6D	0x6D ('m')
17		0x43	IE Identifier = 0x01 (Text); More Flag = 0x00; Length = 0x03 (03) - (APN Password Element – Start)
18		0x74	0x74 ('t')
19		0x69	0x69 ('i')

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20		0x6D	0x6D ('m')
21	Server Configuration Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D (13)
22		0xBD	0xBD (1 st byte – First Server's IP)
23		0x4F	0x4F (2 nd byte - First Server's IP)
24		0x22	0x22 (3 rd byte - First Server's IP)
25		0XF0	0XF0 (4 th byte - First Server's IP)
26		0x15	0x15 (1 st byte - First Server's Port)
27		0xB3	0xB3 (2 nd byte – First Server's Port)
28		0xBD	0xBD (1 st byte - Second Server's IP)
29		0x4F	0x4F (2 nd byte - Second Server's IP)
30		0x22	0x22 (3 rd byte - Second Server's IP)
31		0XF1	0XF1 (4 th byte - Second Server's IP)
32		0x15	0x15 (1 st byte - Second Server's Port)
33		0xBE	0xBE (2 nd byte - Second Server's Port)
34		0x00	Protocol ID = 0x00 (ACP)
35	Control Byte	0x40	More Flag = 0x00; (Activation) = 0x01(Contracted); Reserved = 0x00 (bits 2-7)
36	Vehicle Descriptor Element	0x16	IE Identifier = 0x00; More Flag = 0x00; Length = 0x16 (22)
37		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
38		0x30	Vehicle Descriptor Flag 2 = 0x30 (SIM Card ID and Auth. Key are Presents)
39		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
40		0x12	0x12
41		0x34	0x34
42		0x56	0x56
43		0x78	0x78
44		0x90	0x90
45		0x12	0x12
46		0x34	0x34
47		0x56	0x56
48		0x78	0x78
49		0x90	0x90
50		0x08	IE Identifier = 0x00 (Binary); More Flag = 0x00; Length = 0x08
51		0xXX	0xXX -> Auth Key
52		0xXX	0xXX -> Auth Key
53		0xXX	0xXX -> Auth Key
54		0xXX	0xXX -> Auth Key
55		0xXX	0xXX -> Auth Key
56		0xXX	0xXX -> Auth Key
57		0xXX	0xXX -> Auth Key

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58		0xXX	0xXX -> Auth Key
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13.4.5.2 Configuration Reply Message (From TCU to SO) (Reply for Activation).

Configuration Reply Message (Reply for Activation/Deactivation Message)
Raw Message (Hexadecimal Format)
0203101D04YYZZ01020000420X01000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x03	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x03 (Configuration Reply Msg. #2)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x1D	Message Length = 0x1D (29)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10	Message Fields	0x00	Reserved = 0x00
11		0x00	More Flag = 0x00; Target Application Id = 0x00
12		0x42	AppFlag1 = 0x01 (Activate application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
13		0x0X	Status Flag1 = 0x0X(Accepted values = 0,2,3);TCU Response Flag = 0x00; Reserved = 0x00
14	Error Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
15		0x00	Error Code = 0x00 (OK)
16	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
17		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
18		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
19		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
20		0x12	0x12
21		0x34	0x34
22		0x56	0x56
23		0x78	0x78
24		0x90	0x90
25		0x12	0x12
26		0x34	0x34

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

27		0x56	0x56
28		0x78	0x78
29		0x90	0x90

13.4.5.3 Configuration TCU Service Activation/ Deactivation Message ACP 245 (From SO to TCU) (Deactivation)

Configuration TCU Service Activation/ Deactivation Message ACP 245 (Deactivation)
Raw Message (Hexadecimal Format)
020A11150000000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x0A	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x0A (Activation/Deactivation Message)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01 (Response is expected)
4		0x15	Message Length = 0x15 (21)
5	APN Configuration Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00 (00)
6	Server Configuration Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00 (00)
7	Control Byte	0x00	More Flag = 0x00; Activation = 0x00 (Uncontracted); Reserved = 0x00 (bits 2-7)
8	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
9		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
10		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
11		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
12		0x12	0x12
13		0x34	0x34
14		0x56	0x56
15		0x78	0x78
16		0x90	0x90
17		0x12	0x12
18		0x34	0x34
19		0x56	0x56
20		0x78	0x78
21		0x90	0x90

13.4.5.4 Configuration Reply Message (From TCU to SO) (Reply for Deactivation)

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

Configuration Reply Message (Reply for Deactivation Message)
Raw Message (Hexadecimal Format)
0203101D04YYZZ01020000820X01000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x02	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x02 (Configuration)
2		0x03	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x03 (Configuration Reply Msg. #2)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x1D	Message Length = 0x1D (29)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10	Message Fields	0x00	Reserved = 0x00
11		0x00	More Flag = 0x00; Target Application Id = 0x00
12		0x82	AppFlag1 = 0x02 (Deactivate Application); Control Flag1 = 0x02 (Vehicle Descriptor is Present)
13		0x0X	Status Flag1 = 0x0X(Accepted values = 0,2,3);TCU Response Flag = 0x00; Reserved = 0x00
14	Error Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
15		0x00	Error Code = 0x00 (OK)
16	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
17		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
18		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
19		0x8A	IE Identifier = 0x02(BCD Coding); More Flag = 0x00; Length = 0x0A
20		0x12	0x12
21		0x34	0x34
22		0x56	0x56
23		0x78	0x78
24		0x90	0x90
25		0x12	0x12
26		0x34	0x34
27		0x56	0x56
28		0x78	0x78
29		0x90	0x90

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

13.5 REMOTE VEHICLE FUNCTION

13.5.1 Blocking request: no version element (shortest message)

13.5.1.1 Remote Vehicle Function Command (From SO to TCU)

Remote Vehicle Function Command			
Raw Message (Hexadecimal Format)			
0602111700018001020D80208A12345678901234567890			

Message Detail (Message Elements)			
Octet	Element	Vale	Detail

1	Header Element	0x06	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x06 (Remote Vehicle Function)
2		0x02	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x02 (Remote Vehicle Function Command)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x17	Message Length = 0x17 (23)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Control Function Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
7		0x80	Entity ID = 0x80 (128-Blocking)
8	Function Command Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
9		0x02	Function Command or Status = 0x02 (Start)
10	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
11		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
12		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
13		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
14		0x12	0x12
15		0x34	0x34
16		0x56	0x56
17		0x78	0x78
18		0x90	0x90
19		0x12	0x12
20		0x34	0x34
21		0x56	0x56
22		0x78	0x78
23		0x90	0x90

If vehicle is not remotely blocked then TCU must, first, send a reply through “Remote Vehicle Function Status” and when blocking has occurred send a “Theft Alarm Notification” message.

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If vehicle is already remotely blocked, TCU just sends a reply thru “Remote Vehicle Function Status”.

13.5.1.2 Remote Vehicle Function Status (From TCU to SO)

Remote Vehicle Function Status
Raw Message (Hexadecimal Format)
0603101D04YYZZ01020180010201000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x06	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x06 (Remote Vehicle Function)
2		0x03	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x03 (Remote Vehicle function Status)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x1D	Message Length = 0x1D (29)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10	Control Function Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
11		0x80	Entity ID = 0x80 (128-Blocking)
12	Function Command Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
13		0x02	Function Command or Status = 0x02 (Started)
14	Error Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
15		0x00	Error Code = 0x00 (OK)
16	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
17		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
18		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
19		0x8A	IE Identifier = 0x02(BCD Coding); More Flag = 0x00; Length = 0x0A
20		0x12	0x12
21		0x34	0x34
22		0x56	0x56
23		0x78	0x78
24		0x90	0x90
25		0x12	0x12
26		0x34	0x34

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27	0x56	0x56
28	0x78	0x78
29	0x90	0x90

13.5.2 Unblocking request: no version element (shortest message)

13.5.2.1 Remote Vehicle Function Command (From SO to TCU)

Remote Vehicle Function Command
Raw Message (Hexadecimal Format)
0602111700018001030D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Vale	Detail

1	Header Element	0x06	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x06 (Remote Vehicle Function)
2		0x02	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x02 (Remote Vehicle Function Command)
3		0x11	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x01
4		0x17	Message Length = 0x17 (23)
5	Version Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
6	Control Function Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
7		0x80	Entity ID = 0x80 (128-Blocking)
8	Function Command Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
9		0x03	Function Command or Status = 0x03 (Stop/Disable)
10	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
11		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
12		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
13		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
14		0x12	0x12
15		0x34	0x34
16		0x56	0x56
17		0x78	0x78
18		0x90	0x90
19		0x12	0x12
20		0x34	0x34
21		0x56	0x56
22		0x78	0x78
23		0x90	0x90

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

13.5.2.2 Remote Vehicle Function Status (From TCU to SO)

Remote Vehicle Function Status
Raw Message (Hexadecimal Format)
0603101D04YYZZ01020180010301000D80208A12345678901234567890

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x06	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x06 (Remote Vehicle Function)
2		0x03	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x03 (Remote Vehicle function Status)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x1D	Message Length = 0x1D (29)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x02	Major Software Release = 0x02
10	Control Function Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
11		0x80	Entity ID = 0x80 (128-Blocking)
12	Function Command Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
13		0x03	Function Command or Status = 0x03 (Disabled)
14	Error Element	0x01	IE Identifier = 0x00; More Flag = 0x00; Length = 0x01
15		0x00	Error Code = 0x00 (OK)
16	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
17		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
18		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
19		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
20		0x12	0x12
21		0x34	0x34
22		0x56	0x56
23		0x78	0x78
24		0x90	0x90
25		0x12	0x12
26		0x34	0x34
27		0x56	0x56
28		0x78	0x78
29		0x90	0x90

DENATRAN: ACP 245 V 1.2.2 – Protocol Specification

13.6 VEHICLE TRACKING SERVICE

13.6.1.1 Vehicle Position Message (From TCU to SO) with Vehicle Location Data (Data type 0x0047 = True)

Vehicle Position Message	
Raw Message (Hexadecimal Format)	
0A02103A04YYZZ01014D6D22F8141311300000F6F2BED0FAF5BA2802E4096F0678800D80208A1234567890123456789008808000010380C01100	

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x0A	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x0A (Vehicle Tracking)
2		0x02	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x02 (Vehicle Position)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x3A	Message Length = 0x3A (58)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0x08	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x01	Major Software Release = 0x01
10	Time Stamp Element	0x4D	Year = 0x13 (19 + 1990 = 2009); Month of Year = 0x01
11		0x6D	Month of Year = 0x01 (0101 = 5 = May); Day of Month = 0x16 (10110 = 22 = Day); Hour of Day = 0x01;
12		0x22	Hour of Day = 0x02 (10010 = 18h) ; Minutes = 0x02
13		0xF8	Minutes = 0x03 (001011 = 11 min); Seconds = 0x38 (111000 = 56 sec)
14	Location Element	0x14	IE Identifier = 0x00; More Flag = 0x00; Length = 0x14 (20)
15		0x13	IE Identifier = 0x00; More Flag = 0x00; Length = 0x13 (19) (Current GPS Raw Data Element - Start)
16		0x11	IE Identifier = 0x00; More Flag = 0x00; Length = 0x11 (17) (Area Location Code Element - Start)
17		0x30	More Flag = 0x00; AreaLocationStatusFlag1 = 0x30 (011000)
18		0x00	Area Type = 0x00; Location Type Coding = 0x00; Reserved = 0x00
19		0x00	More Flag = 0x00; Time Difference = 0x00
20		0xF6	Longitude = 0xF6 (Byte1)
21		0xF2	Longitude = 0x02 (Byte2)

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22		0xBE	Longitude = 0xBE (Byte3)
23		0xD0	Longitude = 0xD0 (Byte4)
24		0xFA	Latitude = 0xFA (Byte1)
25		0xF5	Latitude = 0xF5 (Byte2)
26		0xBA	Latitude = 0xBA (Byte3)
27		0x28	Latitude = 0x28 (Byte4)
28		0x02	Altitude = 0x02 (Byte1)
29		0xE4	Altitude = 0xE4 (Byte2)
30		0x09	Position Uncertainty Estimate = 0x04; K/HDOP = 0x01
31		0x6F	Heading Uncertainty Estimate = 0x03; Heading = 0x0F
32		0x06	Reserved = 0x00; Distance Flag = 0x01; Time Flag = 0x02
33		0x78	Velocity = 0x78 (120 km/h) (Area Location Code Element - End)
34		0x80	Number of Satellites = 0x08; Reserved = 0x00
35	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
36		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
37		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
38		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
39		0x12	0x12
40		0x34	0x34
41		0x56	0x56
42		0x78	0x78
43		0x90	0x90
44		0x12	0x12
45		0x34	0x34
46		0x56	0x56
47		0x78	0x78
48		0x90	0x90
49	Break Down Status Element	0x08	IE Identifier = 0x00; More Flag = 0x00; Length = 0x08
50		0x80	More Flag = 0x01; Break Down Source = 0x00 (First Flag)
51		0x80	More Flag = 0x01; Break Down Source = 0x00 (Second Flag)
52		0x00	More Flag = 0x00; Break Down Source = 0x00 (Third Flag)
53		0x01	More Flag = 0x00; Break Down Sensor = 0x01 (Enable Break Down Data to Show Status)
54		0x03	IE Identifier = 0x00; More Flag = 0x00; Length = 0x03
55		0x80	Break Down Data = 0x80
56		0xC0	Break Down Data = 0xC0 (Ignition On)

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57		0x11	Break Down Data = 0x11 (Main battery reconnected and Block off)
58	Information Type Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00

13.6.1.2 Vehicle Position Message (From TCU to SO) without Vehicle Location Data (Data type 0x0047 = False)

Vehicle Position Message			
Raw Message (Hexadecimal Format)			
0A02102604YYZZ01014D6D22F8000D80208A1234567890123456789008808000010380C01100			

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x0A	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x0A (Vehicle Tracking)
2		0x02	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x02 (Vehicle Position)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x26	Message Length = 0x26 (38)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x01	Major Software Release = 0x01
10	Time Stamp Element	0x4D	Year = 0x13 (19 + 1990 = 2009); Month of Year = 0x01
11		0x6D	Month of Year = 0x01 (0101 = 5 = May); Day of Month = 0x16 (10110 = 22 = Day); Hour of Day = 0x01;
12		0x22	Hour of Day = 0x02 (10010 = 18h) ; Minutes = 0x02
13		0xF8	Minutes = 0x03 (001011 = 11 min); Seconds = 0x38 (111000 = 56 sec)
14	Location Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
15	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
16		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
17		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
18		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
19		0x12	0x12
20		0x34	0x34
21		0x56	0x56
22		0x78	0x78
23		0x90	0x90

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24		0x12	0x12
25		0x34	0x34
26		0x56	0x56
27		0x78	0x78
28		0x90	0x90
29	Break Down Status Element	0x08	IE Identifier = 0x00; More Flag = 0x00; Length = 0x08
30		0x80	More Flag = 0x01; Break Down Source = 0x00 (First Flag)
31		0x80	More Flag = 0x01; Break Down Source = 0x00 (Second Flag)
32		0x00	More Flag = 0x00; Break Down Source = 0x00 (Third Flag)
33		0x01	More Flag = 0x00; Break Down Sensor = 0x01 (Enable Break Down Data to Show Status)
34		0x03	IE Identifier = 0x00; More Flag = 0x00; Length = 0x03
35		0x80	Break Down Data = 0x80
36		0xC0	Break Down Data = 0xC0 (Ignition On)
37		0x11	Break Down Data = 0x11 (Main battery reconnected and Block off)
38	Information Type Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00

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13.7 THEFT ALARM

13.7.1.1 Keep Alive (from TCU to SO)

Keep Alive Message	
Raw Message (Hexadecimal Format)	
0B0410120D80208A12345678901234567890	

Message Detail (Message Elements)		
Octet	Element	Detail

1	Header Element	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x0B (Theft Alarm)
2		Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x04 (Keep Alive)
3		Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		Message Length = 0x12 (18)
5	Vehicle Descriptor Element	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
6		Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
7		Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
8		IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
9		0x12
10		0x34
11		0x56
12		0x78
13		0x90
14		0x12
15		0x34
16		0x56
17		0x78
18		0x90

13.7.1.2 Theft Alarm Notification (From TCU to SO) of "Ignition ON" with Vehicle Location Data (Data type 0x0047 = True)

Vehicle Position Message	
Raw Message (Hexadecimal Format)	
0B01103A04YYZZ01014D6D22F8141311300000F6F2BED0FAF5BA2802E4096F0678800D80208A123456789012345678900880C000010380C01100	

Message Detail (Message Elements)			
Octet	Element	Value	Detail

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1	Header Element	0x0B	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x0B(Theft Alarm)
2		0x01	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x01 (Theft Alarm Notification)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x3A	Message Length = 0x3A (58)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x01	Major Software Release = 0x01
10	Time Stamp Element	0x4D	Year = 0x13 (19 + 1990 = 2009); Month of Year = 0x01
11		0x6D	Month of Year = 0x01 (0101 = 5 = May); Day of Month = 0x16 (10110 = 22 = Day); Hour of Day = 0x01;
12		0x22	Hour of Day = 0x02 (10010 = 18h) ; Minutes = 0x02
13		0xF8	Minutes = 0x03 (001011 = 11 min); Seconds = 0x38 (111000 = 56 sec)
14	Location Element	0x14	IE Identifier = 0x00; More Flag = 0x00; Length = 0x14 (20)
15		0x13	IE Identifier = 0x00; More Flag = 0x00; Length = 0x13 (19) (Current GPS Raw Data Element - Start)
16		0x11	IE Identifier = 0x00; More Flag = 0x00; Length = 0x11 (17) (Area Location Code Element - Start)
17		0x30	More Flag = 0x00; AreaLocationStatusFlag1 = 0x30 (011000)
18		0x00	Area Type = 0x00; Location Type Coding = 0x00; Reserved = 0x00
19		0x00	More Flag = 0x00; Time Difference = 0x00
20		0xF6	Longitude = 0xF6 (Byte1)
21		0xF2	Longitude = 0x02 (Byte2)
22		0xBE	Longitude = 0xBE (Byte3)
23		0xD0	Longitude = 0xD0 (Byte4)
24		0xFA	Latitude = 0xFA (Byte1)
25		0xF5	Latitude = 0xF5 (Byte2)
26		0xBA	Latitude = 0xBA (Byte3)
27		0x28	Latitude = 0x28 (Byte4)
28		0x02	Altitude = 0x02 (Byte1)
29		0xE4	Altitude = 0xE4 (Byte2)
30		0x09	Position Uncertainty Estimate = 0x04; K/HDOP = 0x01
31		0x6F	Heading Uncertainty Estimate = 0x03; Heading = 0x0F
32		0x06	Reserved = 0x00; Distance Flag = 0x01; Time Flag =

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		0x02	
33		0x78	Velocity = 0x78 (120 km/h) (Area Location Code Element - End)
34		0x80	Number of Satellites = 0x08; Reserved = 0x00
35	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
36		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
37		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
38		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
39		0x12	0x12
40		0x34	0x34
41		0x56	0x56
42		0x78	0x78
43		0x90	0x90
44		0x12	0x12
45		0x34	0x34
46		0x56	0x56
47		0x78	0x78
48		0x90	0x90
49	Break Down Status Element	0x08	IE Identifier = 0x00; More Flag = 0x00; Length = 0x08
50		0x80	More Flag = 0x01; Break Down Source = 0x00 (First Flag) (<i>More Flag=1</i>)
51		0xC0	More Flag = 0x01; Break Down Source = 1000000 (Second Flag) (<i>More Flag=1, Ignition ON=1</i>)
52		0x00	More Flag = 0x00; Break Down Source = 0x00 (Third Flag)
53		0x01	More Flag = 0x00; Break Down Sensor = 0x01 (Enable Break Down Data to Show Status)
54		0x03	IE Identifier = 0x00; More Flag = 0x00; Length = 0x03
55		0x80	Break Down Data = 0x80
56		0xC0	Break Down Data = 0xC0 (<i>Ignition On</i>)
57		0x11	Break Down Data = 0x11 (<i>Main battery reconnected and Block off</i>)
58	Information Type Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00

13.7.1.3 Theft Alarm Notification (From TCU to SO) of "Ignition ON" without Vehicle Location Data (Data type 0x0047 = False).

Vehicle Position Message	
Raw Message (Hexadecimal Format)	
0B01102604YYZZ01014D6D22F8000D80208A123456789012345678900880C000010380C01100	
Message Detail (Message Elements)	

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Octet	Element	Value	Detail
1	Header Element	0x0B	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x0B(Theft Alarm)
2		0x01	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x01 (Theft Alarm Notification)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x26	Message Length = 0x26 (38)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x01	Major Software Release = 0x01
10	Time Stamp Element	0x4D	Year = 0x13 (19 + 1990 = 2009); Month of Year = 0x01
11		0x6D	Month of Year = 0x01 (0101 = 5 = May); Day of Month = 0x16 (10110 = 22 = Day); Hour of Day = 0x01;
12		0x22	Hour of Day = 0x02 (10010 = 18h) ; Minutes = 0x02
13		0xF8	Minutes = 0x03 (001011 = 11 min); Seconds = 0x38 (111000 = 56 sec)
14	Location Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
15	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
16		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
17		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
18		0x8A	IE Identifier = 0x02(BCD Coding); More Flag = 0x00; Length = 0x0A
19		0x12	0x12
20		0x34	0x34
21		0x56	0x56
22		0x78	0x78
23		0x90	0x90
24		0x12	0x12
25		0x34	0x34
26		0x56	0x56
27		0x78	0x78
28		0x90	0x90
29	Break Down Status Element	0x08	IE Identifier = 0x00; More Flag = 0x00; Length = 0x08
30		0x80	More Flag = 0x01; Break Down Source = 0x00 (First Flag)
31		0xC0	More Flag = 0x01; Break Down Source = 1000000 (Second Flag) (<i>More Flag=1, Ignition ON=1</i>)
32		0x00	More Flag = 0x00; Break Down Source = 0x00 (Third

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			Flag)
33		0x01	More Flag = 0x00; Break Down Sensor = 0x01 (Enable Break Down Data to Show Status)
34		0x03	IE Identifier = 0x00; More Flag = 0x00; Length = 0x03
35		0x80	Break Down Data = 0x80
36		0xC0	Break Down Data = 0xC0 (Ignition On)
37		0x11	Break Down Data = 0x11 (Main battery reconnected and Block off)
38	Information Type Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00

13.7.1.4 Theft Alarm in event mode Vehicle Position Message (From TCU to SO) with Vehicle Location Data (Data type 0x0047 = True)

Vehicle Position Message	
Raw Message (Hexadecimal Format)	
0B03103A04YYZZ01014D6D22F8141311300000F6F2BED0FAF5BA2802E4096F0678800D80208A1234567890123456789008808000010380C01100	

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x0B	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x0B(Theft Alarm)
2		0x03	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x03 (Theft Alarm in event mode)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x3A	Message Length = 0x3A (58)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x01	Major Software Release = 0x01
10	Time Stamp Element	0x4D	Year = 0x13 (19 + 1990 = 2009); Month of Year = 0x01
11		0x6D	Month of Year = 0x01 (0101 = 5 = May); Day of Month = 0x16 (10110 = 22 = Day); Hour of Day = 0x01;
12		0x22	Hour of Day = 0x02 (10010 = 18h) ; Minutes = 0x02
13		0xF8	Minutes = 0x03 (001011 = 11 min); Seconds = 0x38 (111000 = 56 sec)
14	Location Element	0x14	IE Identifier = 0x00; More Flag = 0x00; Length = 0x14 (20)
15		0x13	IE Identifier = 0x00; More Flag = 0x00; Length = 0x13 (19) (Current GPS Raw Data Element - Start)

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16		0x11	IE Identifier = 0x00; More Flag = 0x00; Length = 0x11 (17) (Area Location Code Element - Start)
17		0x30	More Flag = 0x00; AreaLocationStatusFlag1 = 0x30 (011000)
18		0x00	Area Type = 0x00; Location Type Coding = 0x00; Reserved = 0x00
19		0x00	More Flag = 0x00; Time Difference = 0x00
20		0xF6	Longitude = 0xF6 (Byte1)
21		0xF2	Longitude = 0x02 (Byte2)
22		0xBE	Longitude = 0xBE (Byte3)
23		0xD0	Longitude = 0xD0 (Byte4)
24		0xFA	Latitude = 0xFA (Byte1)
25		0xF5	Latitude = 0xF5 (Byte2)
26		0xBA	Latitude = 0xBA (Byte3)
27		0x28	Latitude = 0x28 (Byte4)
28		0x02	Altitude = 0x02 (Byte1)
29		0xE4	Altitude = 0xE4 (Byte2)
30		0x09	Position Uncertainty Estimate = 0x04; K/HDOP = 0x01
31		0x6F	Heading Uncertainty Estimate = 0x03; Heading = 0x0F
32		0x06	Reserved = 0x00; Distance Flag = 0x01; Time Flag = 0x02
33		0x78	Velocity = 0x78 (120 km/h) (Area Location Code Element - End)
34		0x80	Number of Satellites = 0x08; Reserved = 0x00
35	Vehicle Descriptor Element	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D
36		0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
37		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
38		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
39		0x12	0x12
40		0x34	0x34
41		0x56	0x56
42		0x78	0x78
43		0x90	0x90
44		0x12	0x12
45		0x34	0x34
46		0x56	0x56
47		0x78	0x78
48		0x90	0x90
49	Break Down Status Element	0x08	IE Identifier = 0x00; More Flag = 0x00; Length = 0x08
50		0x80	More Flag = 0x01; Break Down Source = 0x00 (First Flag)

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51		0x80	More Flag = 0x01; Break Down Source = 0x00 (Second Flag)
52		0x00	More Flag = 0x00; Break Down Source = 0x00 (Third Flag)
53		0x01	More Flag = 0x00; Break Down Sensor = 0x01 (Enable Break Down Data to Show Status)
54		0x03	IE Identifier = 0x00; More Flag = 0x00; Length = 0x03
55		0x80	Break Down Data = 0x80
56		0xC0	Break Down Data = 0xC0 (Ignition On)
57		0x11	Break Down Data = 0x11 (Main battery reconnected and Block off)
58	Information Type Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00

13.7.1.5 Theft Alarm in event mode Vehicle Position Message (From TCU to SO) without Vehicle Location Data (Data type 0x0047 = False)

Vehicle Position Message	
Raw Message (Hexadecimal Format)	
0B03102604YYZZ01014D6D22F8000D80208A1234567890123456789008808000010380C01100	

Message Detail (Message Elements)			
Octet	Element	Value	Detail

1	Header Element	0x0B	Reserved = 0x00; Non Standard Flag = 0x00; App Id = 0x0B (Theft Alarm)
2		0x03	Reserved = 0x00; Private Flag = 0x00; Test Flag = 0x00; Message Type = 0x03 (Theft Alarm in event mode)
3		0x10	Version Flag = 0x00; Version = 0x01; Message Control Flag = 0x00
4		0x26	Message Length = 0x26 (38)
5	Version Element	0x04	IE Identifier = 0x00; More Flag = 0x00; Length = 0x04
6		0xYY	Car Manufacturer ID = 0xYY (0xYY is defined in the Table 3 – Version Element: Car Manufacturer IDs.)
7		0xZZ	TCU Manufacturer ID = 0xZZ (0xZZ is defined in the Table 4 – Version Element: TCU Manufacturer IDs.)
8		0x01	Major Hardware Release = 0x01
9		0x01	Major Software Release = 0x01
10	Time Stamp Element	0x4D	Year = 0x13 (19 + 1990 = 2009); Month of Year = 0x01
11		0x6D	Month of Year = 0x01 (0101 = 5 = May); Day of Month = 0x16 (10110 = 22 = Day); Hour of Day = 0x01;
12		0x22	Hour of Day = 0x02 (10010 = 18h) ; Minutes = 0x02
13		0xF8	Minutes = 0x03 (001011 = 11 min); Seconds = 0x38 (111000 = 56 sec)
14	Location Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00
15	Vehicle Descriptor	0x0D	IE Identifier = 0x00; More Flag = 0x00; Length = 0x0D

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16	Element	0x80	Vehicle Descriptor Flag 1 = 0x80 (has more Flag)
17		0x20	Vehicle Descriptor Flag 2 = 0x20 (SIM Card ID is Present)
18		0x8A	IE Identifier = 0x02 (BCD Coding); More Flag = 0x00; Length = 0x0A
19		0x12	0x12
20		0x34	0x34
21		0x56	0x56
22		0x78	0x78
23		0x90	0x90
24		0x12	0x12
25		0x34	0x34
26		0x56	0x56
27		0x78	0x78
28		0x90	0x90
29	Break Down Status Element	0x08	IE Identifier = 0x00; More Flag = 0x00; Length = 0x08
30		0x80	More Flag = 0x01; Break Down Source = 0x00 (First Flag)
31		0x80	More Flag = 0x01; Break Down Source = 0x00 (Second Flag)
32		0x00	More Flag = 0x00; Break Down Source = 0x00 (Third Flag)
33		0x01	More Flag = 0x00; Break Down Sensor = 0x01 (Enable Break Down Data to Show Status)
34		0x03	IE Identifier = 0x00; More Flag = 0x00; Length = 0x03
35		0x80	Break Down Data = 0x80
36		0xC0	Break Down Data = 0xC0 (Ignition On)
37		0x11	Break Down Data = 0x11 (Main battery reconnected and Block off)
38	Information Type Element	0x00	IE Identifier = 0x00; More Flag = 0x00; Length = 0x00

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13.7.1.6 Extended Version Example.

HEADER DESCRIPTION

Octet / Bit	0	1	2	3	4	5	6	7
1	Reserved Set to 0	Private flag	Application ID					
2	Reserved Set to 0	Private flag	Test Flag	Message Type				
3	Version Flag = 1	Version = 7			Message Control Flag			
4	More flag	Extended Version = 7					Message Priority flag = 0	
5, 6	Message Length							

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14 Appendix III – Communication Flow – Use Rules

This attachment has been created to define the mandatory conditions of ACP 245 operations by a standard communication flow between SO (Service Operator) and TCU. All implementation of ACP 245 must proceed with these rules described below.

14.1 ACP 245 Message Replies

14.1.1 Defined messages in ACP 245

Reply messages must have the same “Application ID” from the original message with the proper error code in “Error Element”, according to Application ID used.

The “Bit 3” of “Message Control Flag” if set to “1”, it will always request a reply from the receiver.

If the SO is expecting a reply message, is mandatory that the SO receive it or wait for time expiration (timeout) before sending a new message.

Start Message	Reply Message <i>(always if required)</i>	Flow
Provision Update Message #1	Provision Reply Message #1	SO -> TCU
Configuration Update Message #2 ACP 245	Configuration Reply Message	SO -> TCU
	Configuration Reply Message #2 ACP 245	SO -> TCU
Configuration TCU Service Activation/Deactivation Message ACP 245	Configuration Reply Message	SO -> TCU
Remote Vehicle Function Command	Remote Vehicle Function Status	SO -> TCU
Tracking - Vehicle Position Message	Vehicle Position Reply	TCU -> SO
Theft Alarm Notification	Theft Alarm Reply	TCU -> SO
Theft Alarm - Vehicle Position Message	Theft Alarm Reply	TCU -> SO

14.1.2 Messages present in the ACP 3.1.0.2 and not defined in ACP 245

There are only two options allowed:

- Ignore it and do not send a reply message with error code;
- OR
- Follow the ACP 3.1.0.2 procedures.

14.1.3 Invalid Messages

Ignore it and do not send a reply message with error code.

14.2 The transport layer defined for ACP 245 is the TCP - IP.

14.3 Opening a Connection

14.3.1 Every connection always started by TCU.

14.3.2 The following elements must be present in the first message from TCU:

- 1) “Vehicle Descriptor Element” with ICCID;
- 2) Full “Version Element”;

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14.4 All messages from the TCU and from the SO must contain the *Vehicle Descriptor Element* with ICCID.

14.5 In the message Keep Alive is optional to include *Vehicle Descriptor Element* with ICCID.

14.6 In the messages Provision Update Message#1 and Configuration Update Message #2 ACP 245 the fields *ControlFlag2*, *Start Time*, *End Time* and *Grace Time* are optional and dependent of the TCU manufacturer implementation. The default values of *ControlFlag1* must show these fields as absent (Bit 0=0, Bit 1=0, Bit 2=0 and Bit 3=0).

14.7 When Vehicle Location Function is not active (Data type 0x0047 = false):

- i. Location Element in any message must have length set to 0 (zero).

14.8 At the time of activating the SO service:

- i. In the case of TCU does not send automatically the position history stored, the SO must request it through the message **“Remote Vehicle Function Command”** (*Application*=6, *Message type* = 2) with *Entity ID* =17 and *Function command* =2 (Enable/Start).

14.9 When Vehicle Location Function is active (Data type 0x0047 = true):

- i. With the ignition status "ON", the TCU must record at least-one position as configured by SO with parameter 0x0011 and sent it to the SO as soon as a TCP connection is available or every 24 hours minimum.
- ii. One position must be stored at the every new ignition cycle and sent to the SO as soon as a TCP connection is available or every 24 hours minimum.
- ii. With the ignition status OFF, the TCU must connect with SO, at least once every 24 hours and send the recorded positions that have not been sent and the current position with *timestamp*.

14.10 TCU doesn't accept to STOP tracking with “Function command or Status”=3 (Disable or Stop) and “Entity ID”=1 (Vehicle Tracking). In this case, TCU replies with a “Remote Vehicle Function Status” message with “Error Code=19”. To STOP Tracking, SO must use Data type 0x0011 (Tracking timer) set to 0 (zero).

14.11 When Vehicle Tracking Function is active, Data type 0x0011 (Tracking timer) must be greater than 0 (zero). In this case, TCU sends vehicle position message periodically.

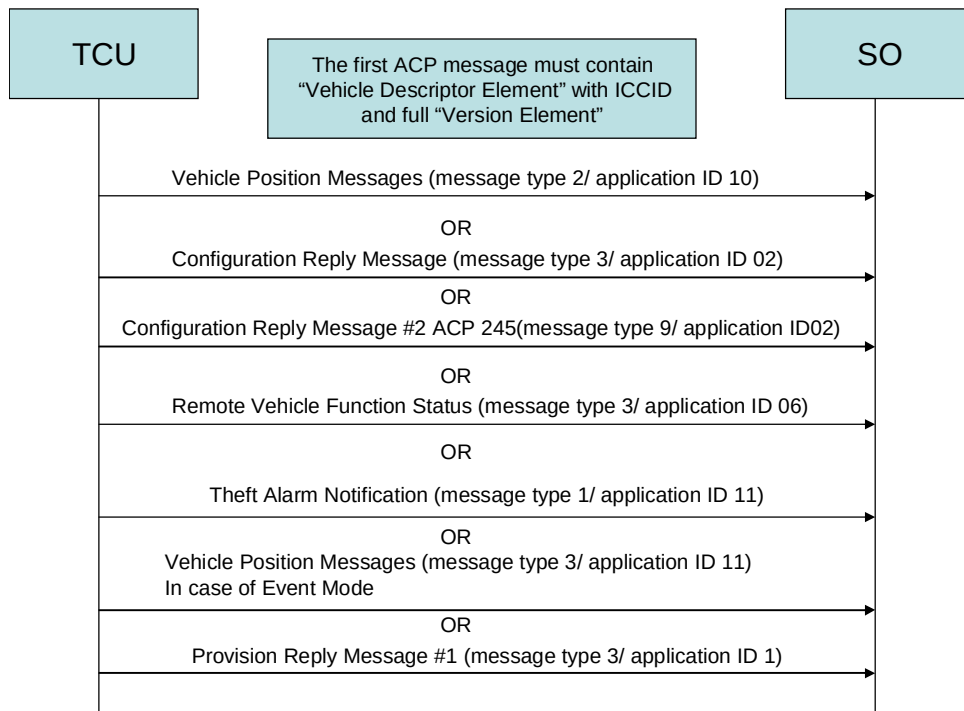
14.12 In the case of SO utilizes “Entity ID”=1 (Vehicle Tracking) and “Transmit Units” assigned to values = 0, 1 or 2, the data type 0x0011 (Tracking timer) is updated with the content of “transmit interval” field up to the limit of 65.535 seconds.

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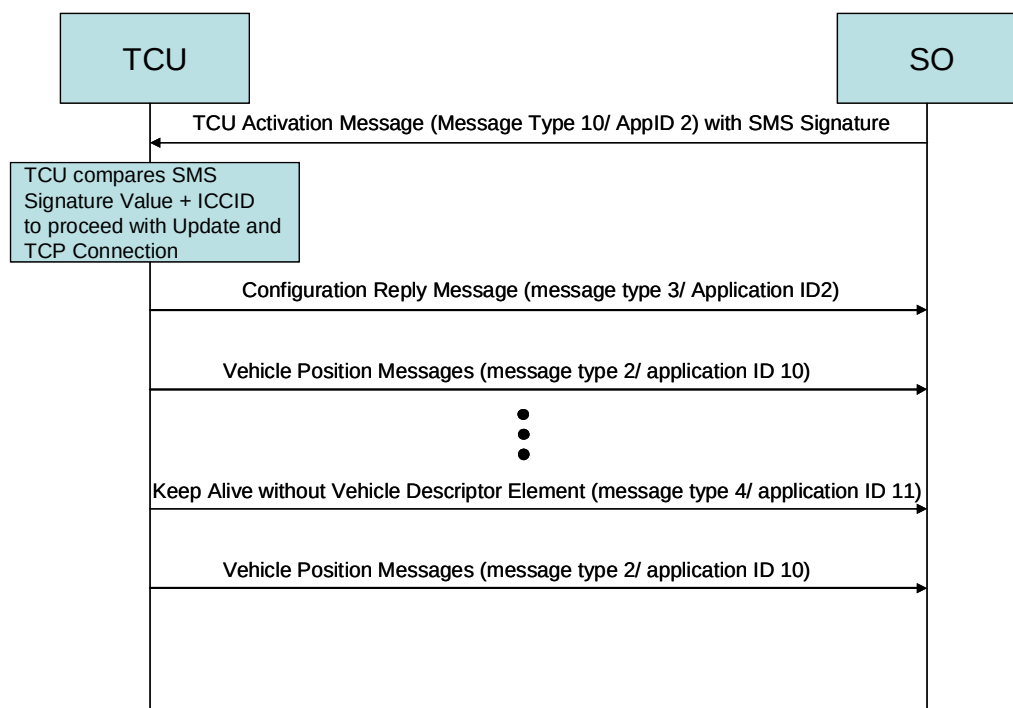
14.13

Messages Flow in ACP 245

14.13.1 TCU Reconnection

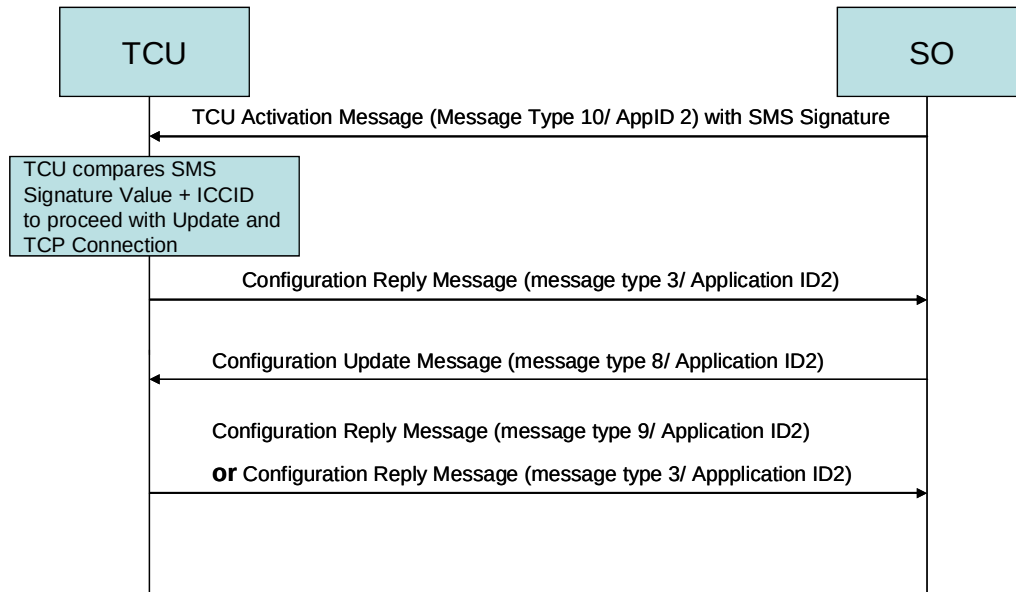


14.13.2 TCU Activation

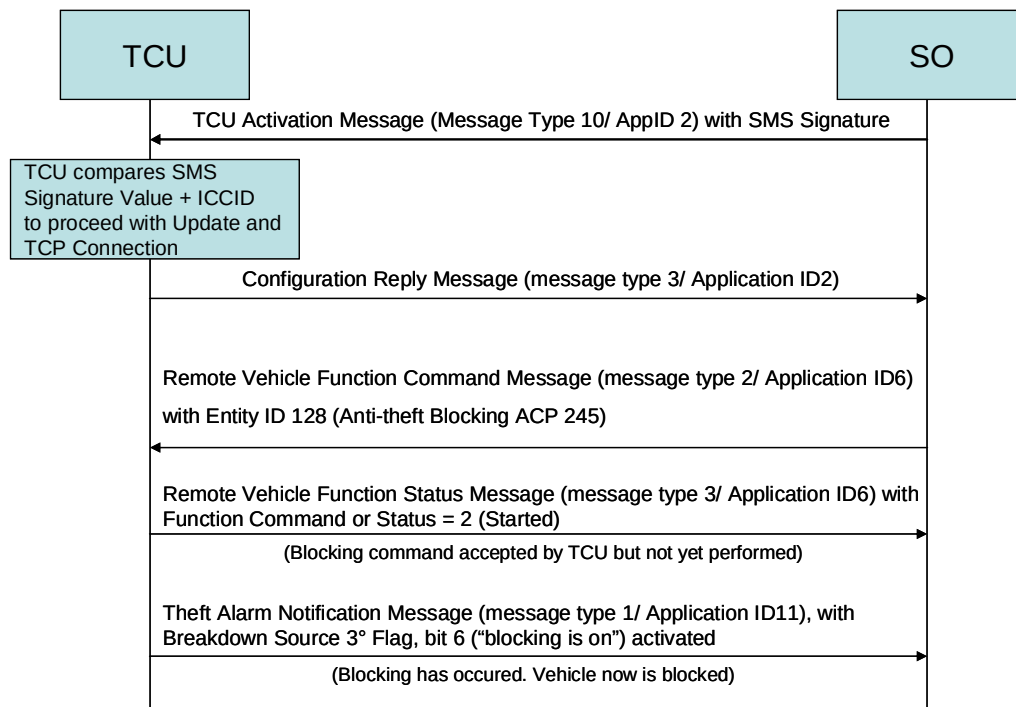


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14.13.3 TCU Activation/ Configuration

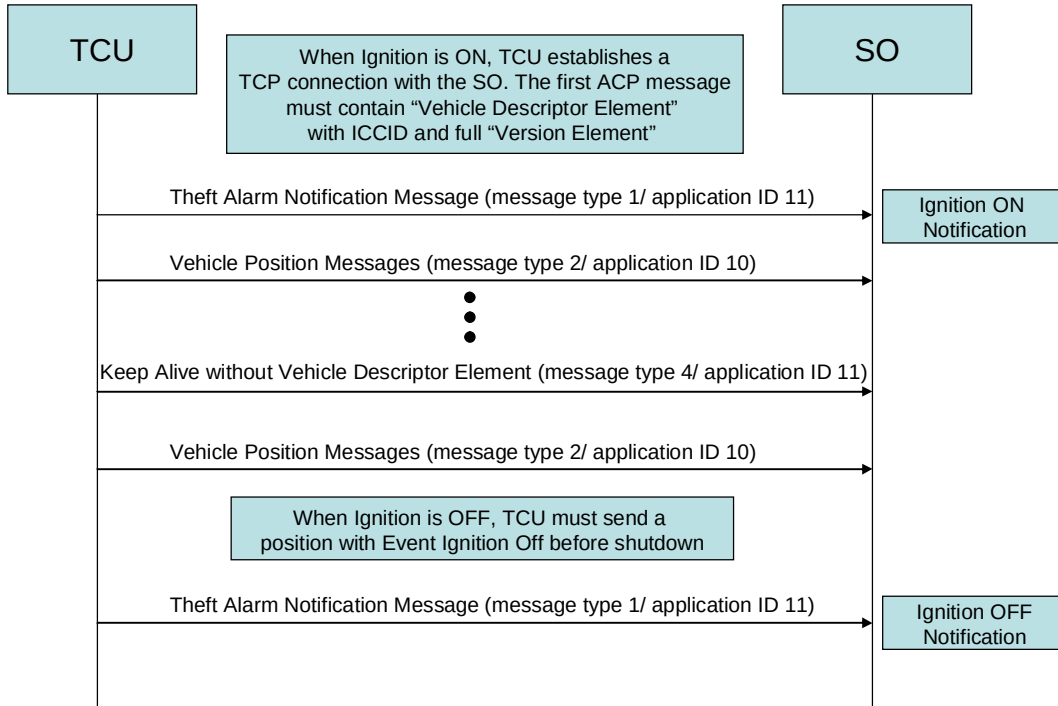


14.13.4 TCU Activation/ Blocking

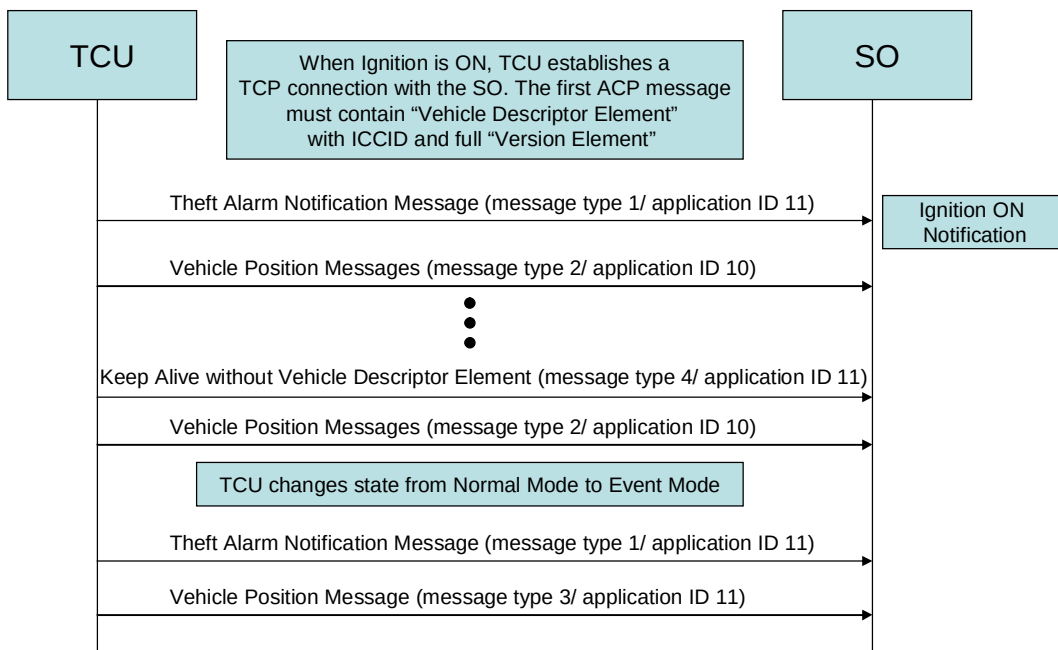


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14.13.5 Ignition On / TCU Tracking Normal Mode / Ignition Off



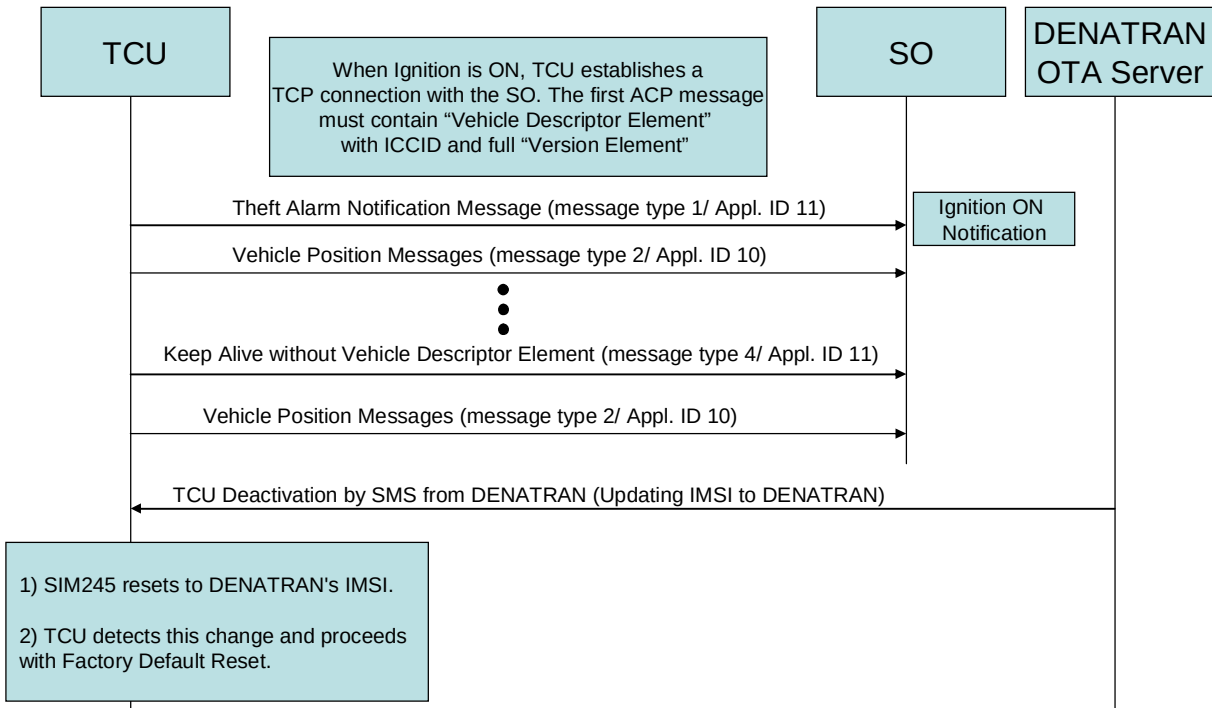
14.13.6 Ignition On / TCU Tracking Normal Mode / Event Mode



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14.13.7 Ignition Start/ TCU Tracking/ Deactivation

14.13.7.1 Deactivation with SIM 245 Generic



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14.13.7.2 Deactivation with SIM 245 Pre Programmed or Branded SIM Card

